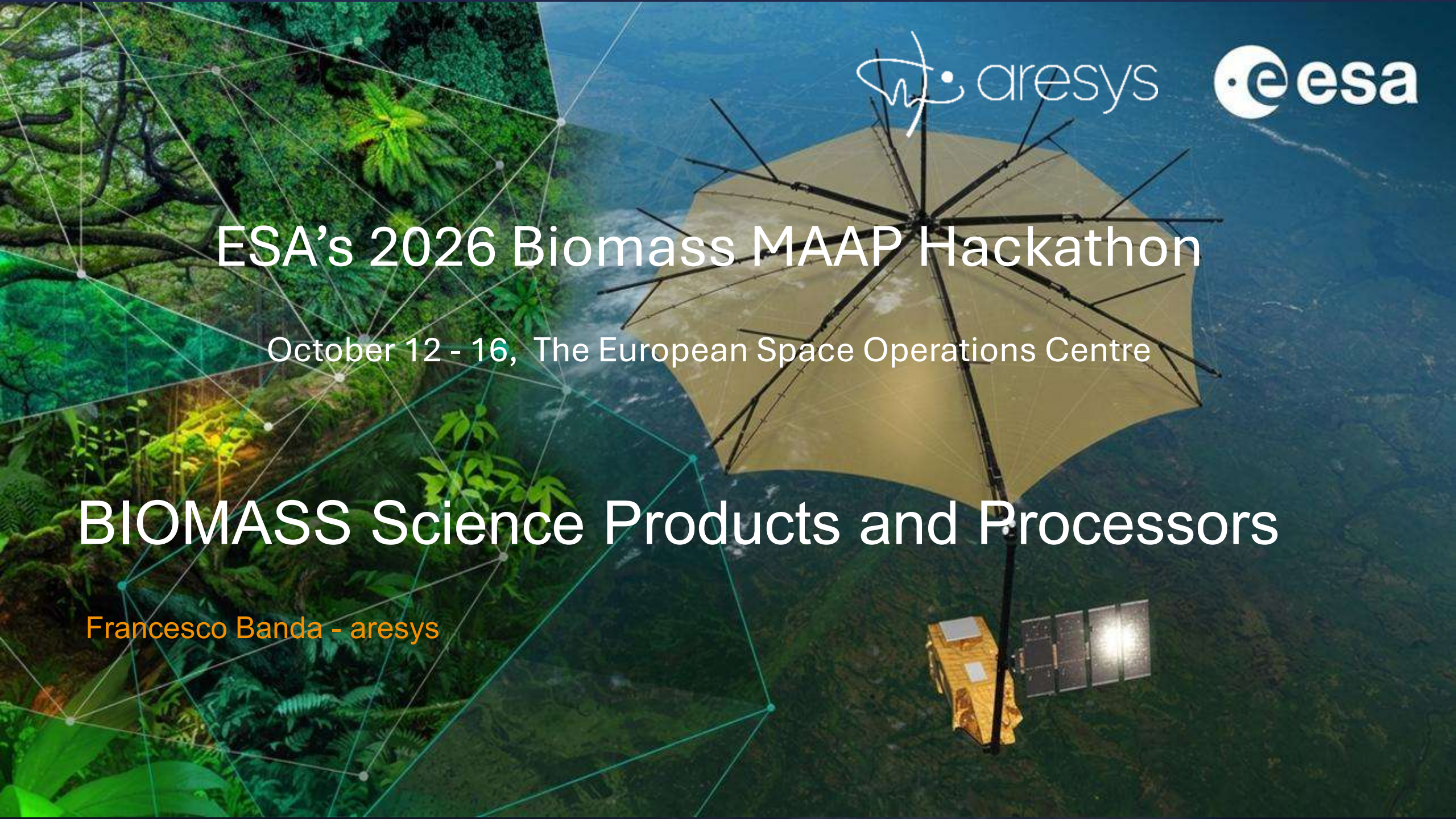


ESA's 2026 Biomass MAAP Hackathon

October 12 - 16, The European Space Operations Centre

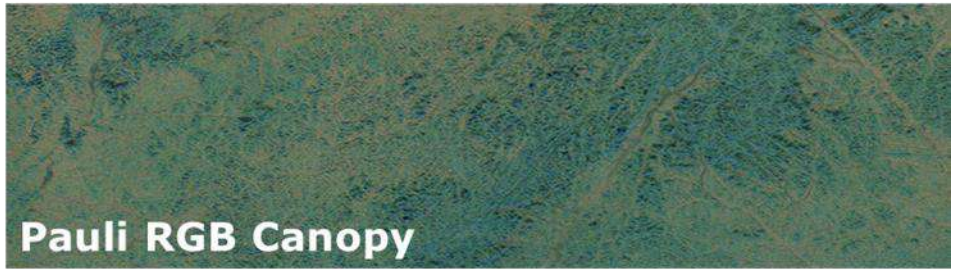
BIOMASS Science Products and Processors

Francesco Banda - aresys

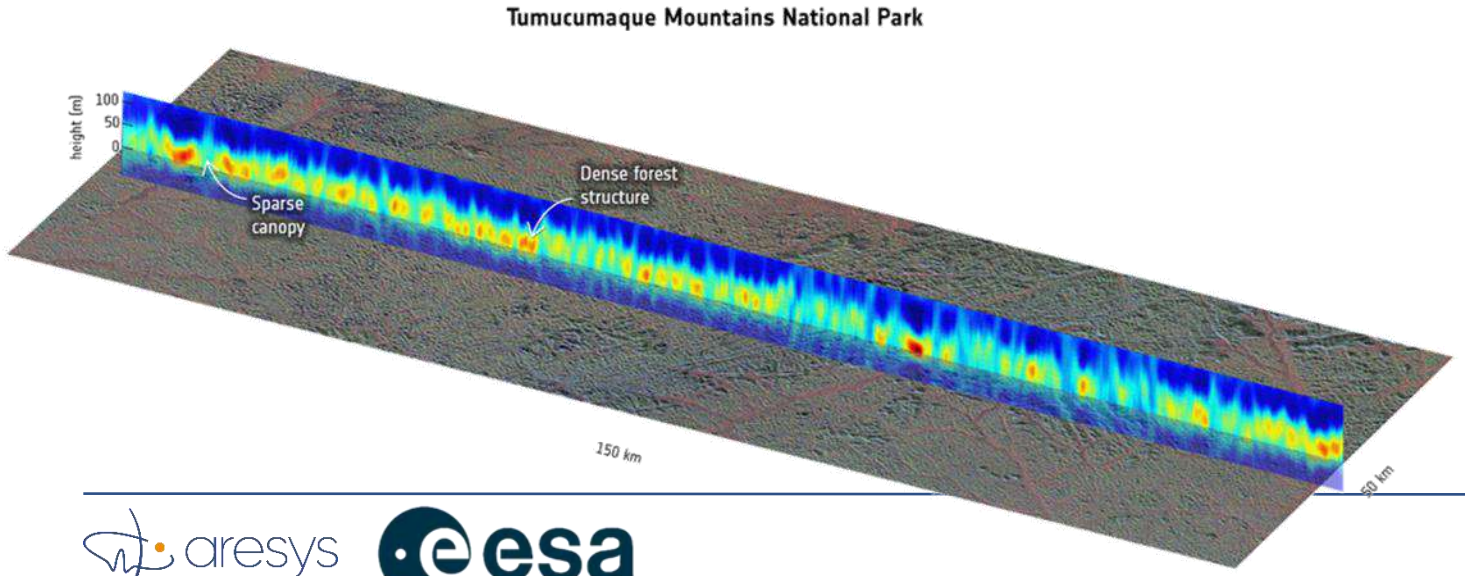
Science and products

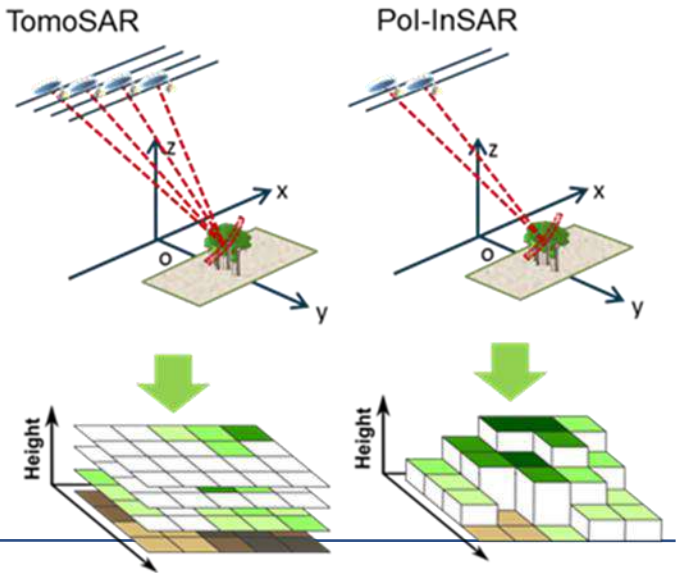
BIOMASS





- ESA's 7th Earth Explorer deployed on 29th April 2025
- **The first P-band SAR ever flown in space**
- **Interferometric and fully-polarimetric**
- Primarily designed for forests (but many other applications!)
- P-Band waves (λ=70 cm) penetrate the vegetation layer down to the underlying terrain, while giving rise to backscattering from trunks and branches, providing sensitivity to the whole forest vertical structure

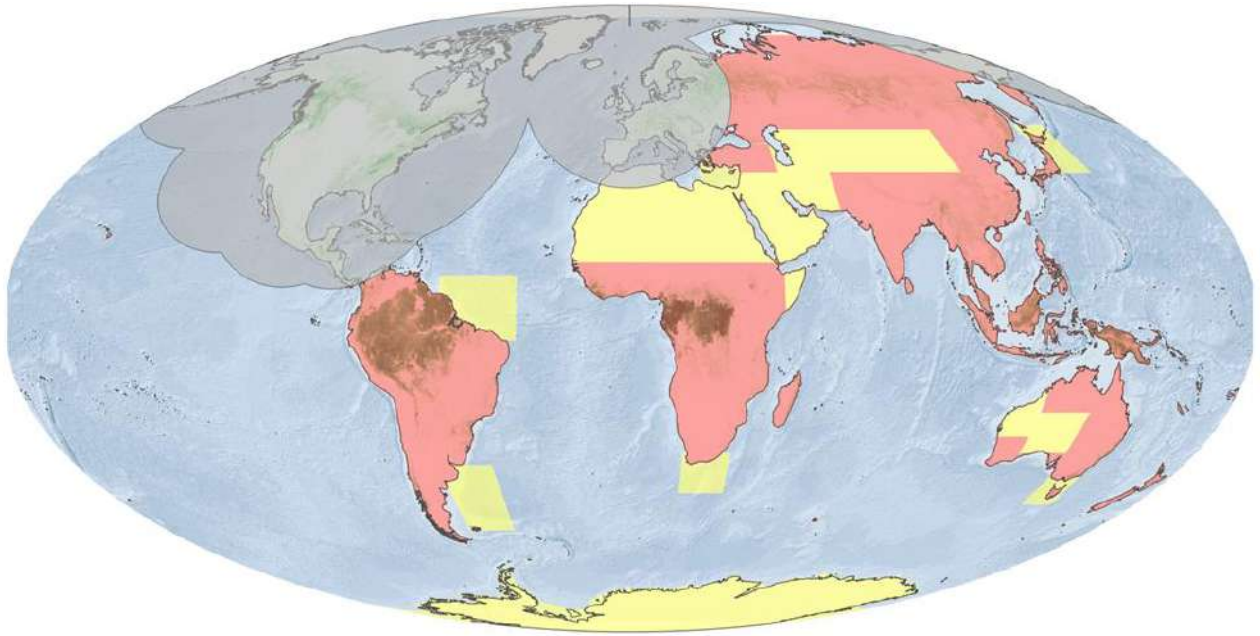


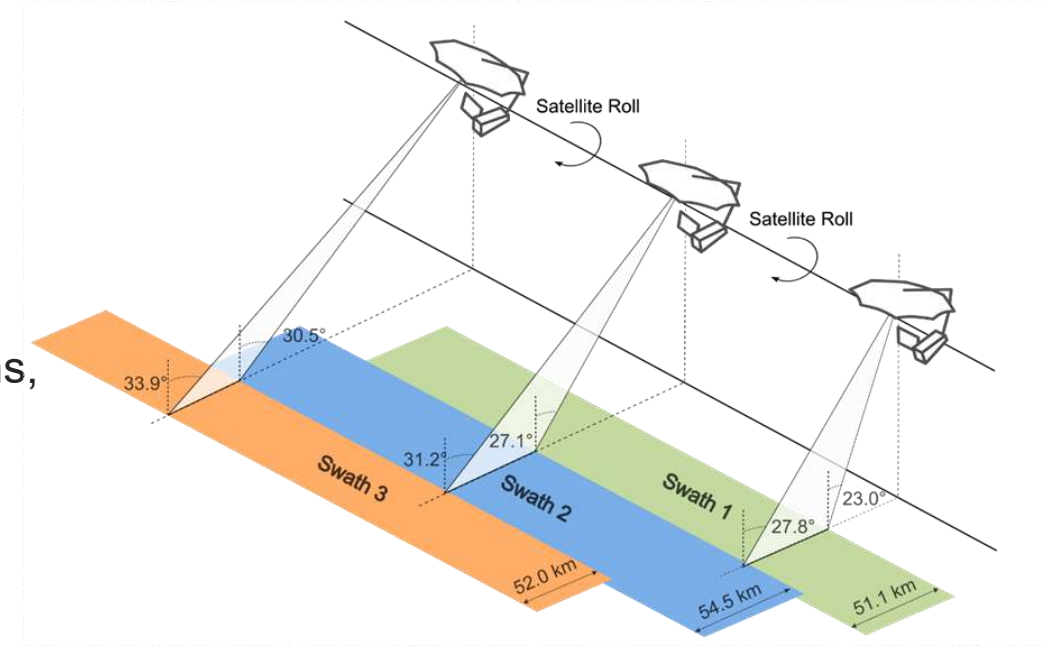


Coverage and phases

BIOMASS phases

- In-Orbit Commissioning Phase
- **Tomographic phase (now): 16 months, nominally 7 acquisitions, alternating swath every 21 days, one global coverage**
- Interferometric phase (spring 2027 till the end): nominally 3 acquisitions, alternating swath every 9 days, about 6 global coverages





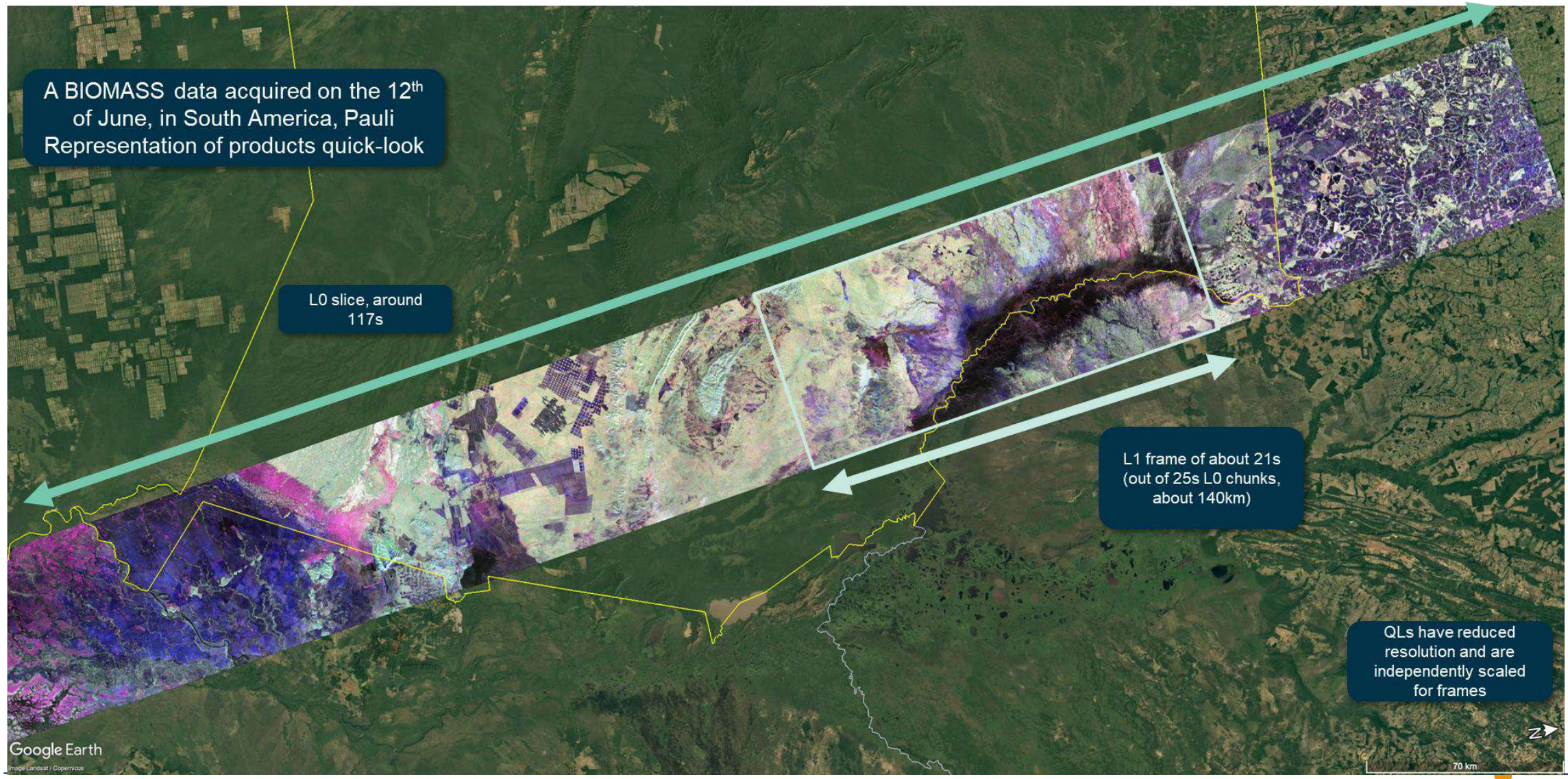
Coverage

- Primary: forested land (red)
- Secondary (yellow areas)
- Exclusion over EU and NA due to Space Objects Tracking Radar

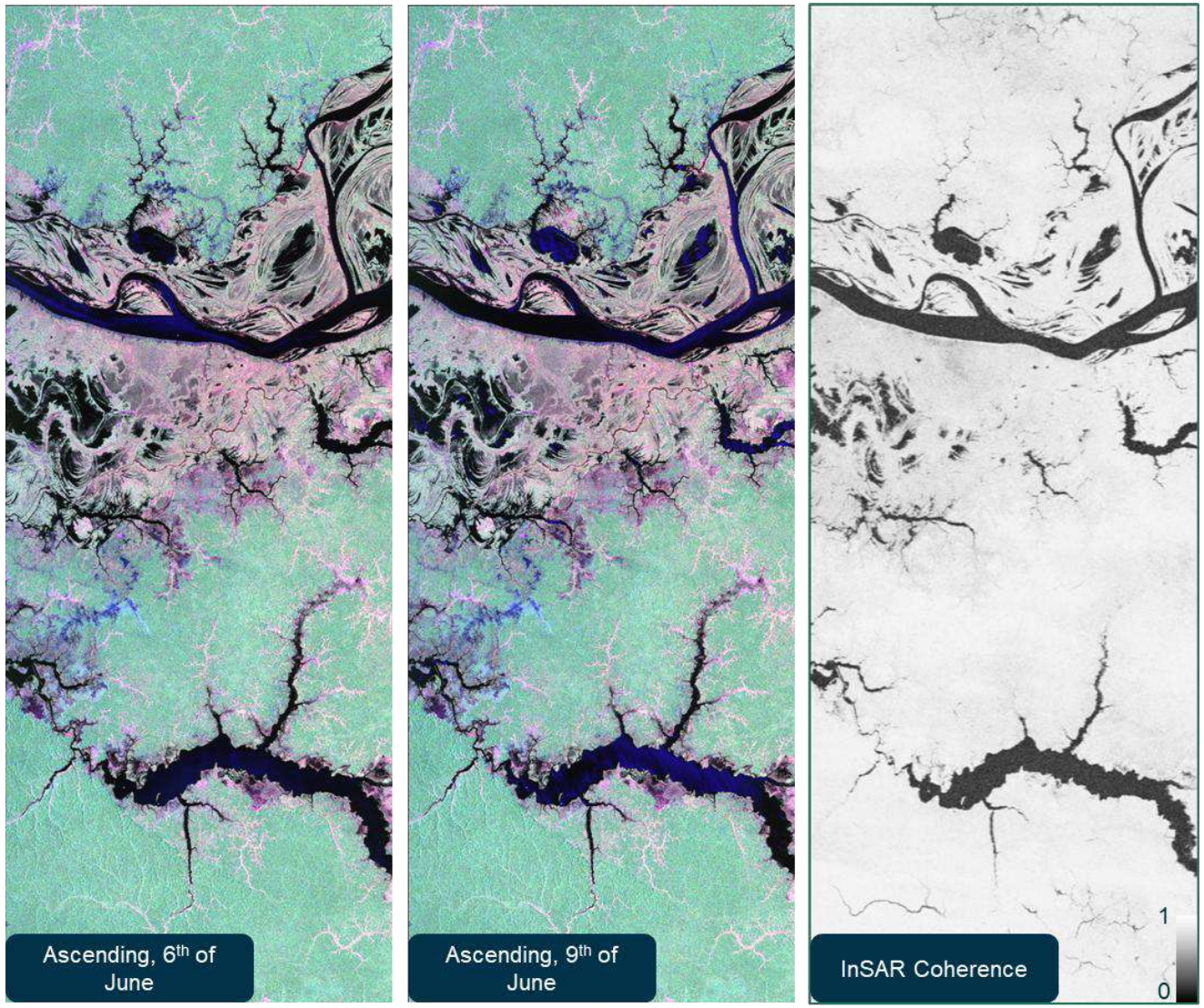
BIOMASS Level-1 Products

Level 1A	Level 1B	Level 1C
Single-look Complex Slant (SCS) data	**Detected Ground Multi-look (DGM) data**	**Stack interferometric product (STA)**
Complex data (abs and phase) in zero Doppler plane and slant-range projection	Detected data in zero Doppler plane and ground-range projection, multilooked	*N* complex data (abs and phase) in zero Doppler plane and slant-range projection, co-registered to common grid and calibrated. Default is N = 3 for INT and N= 7 for TOM phases
Resolution (rg x az [m]): 22,8 x 7,6 to 23,5 x 7,9	Resolution (rg x az [m]): 54,3 x 50,5 to 55,3 x 50,5	Resolution (rg x az [m]): 22,8 x 7,6 to 23,5 x 7,9
Spacing (rg x az [m]): 19,8 x 6,7	Spacing (rg x az [m]): 25 x 25	Spacing (rg x az [m]): 19,8 x 6,7
Looks 1 x 1	Looks 1 x 6	Looks 1 x 1

BIOMASS Level-1 Products



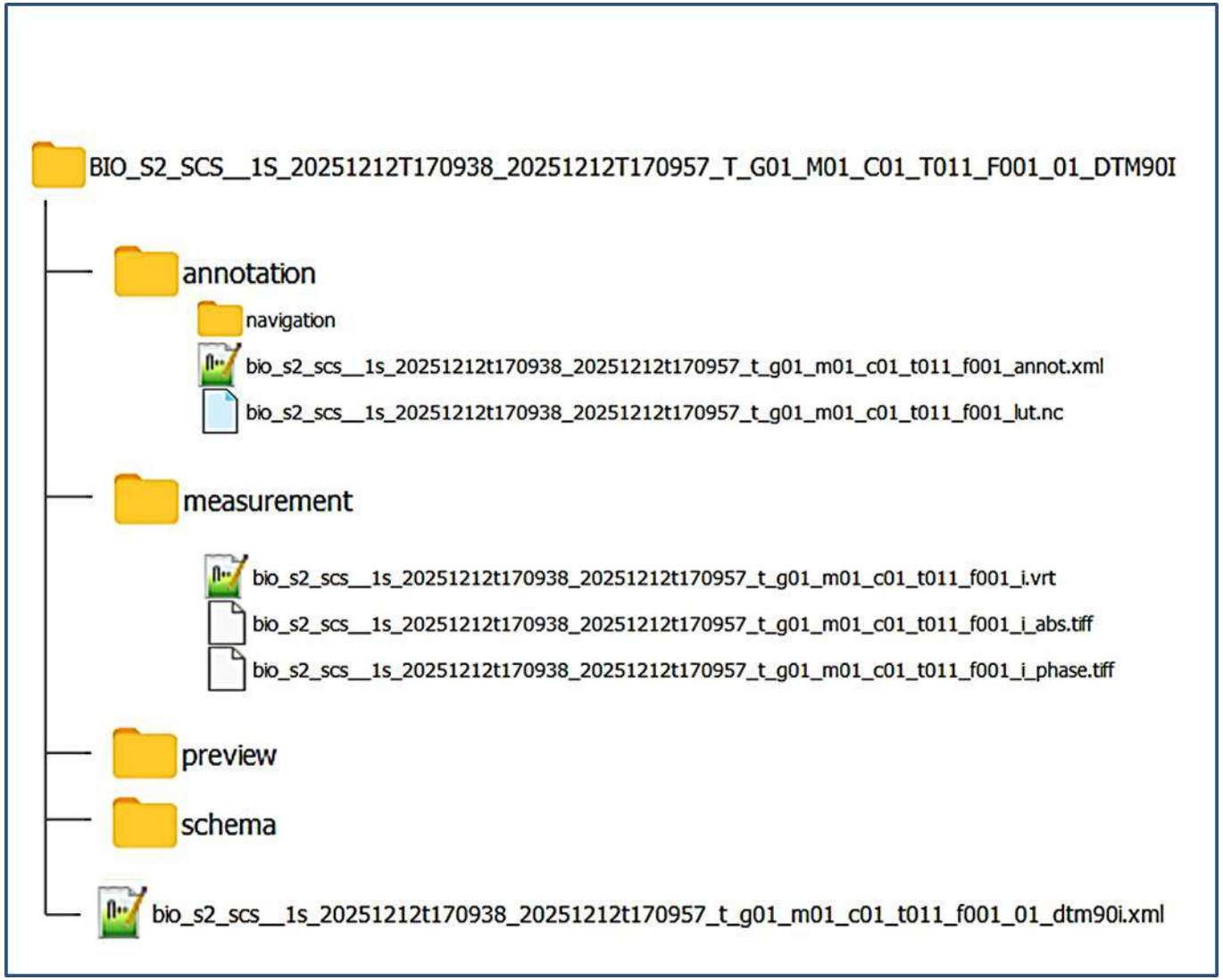
BIOMASS Level-1 Products



- **InSAR coherence** measures the correlation between two complex SAR images
- Determined by system effects, temporal variability and structure
  - magnitude ranges from 0 to 1
  - phase contains information about motion, topography and atmosphere

$$\gamma = \frac{|\langle S_1 S_2^* \rangle|}{\sqrt{\langle |S_1|^2 \rangle \langle |S_2|^2 \rangle}}$$

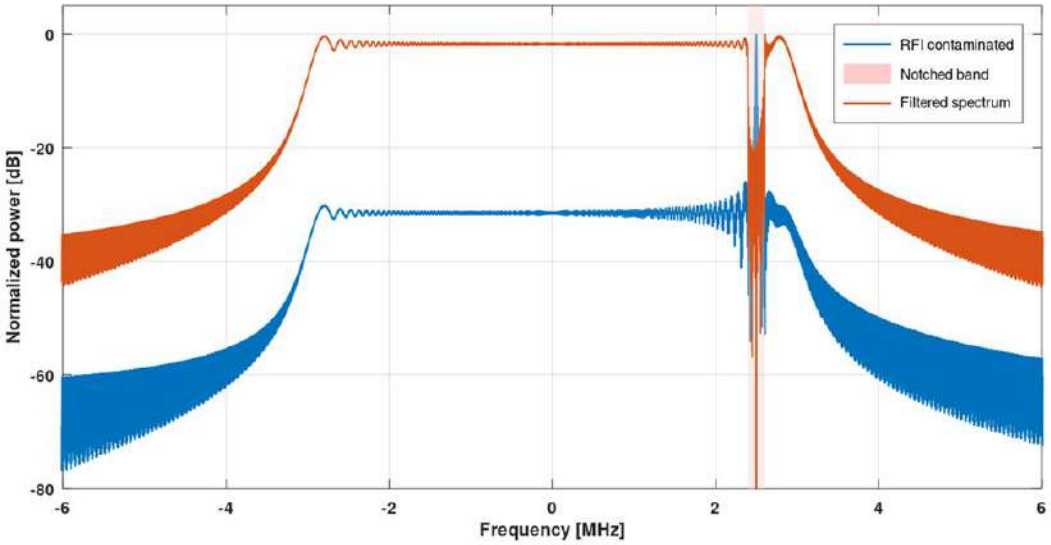
Level-1a Product Format



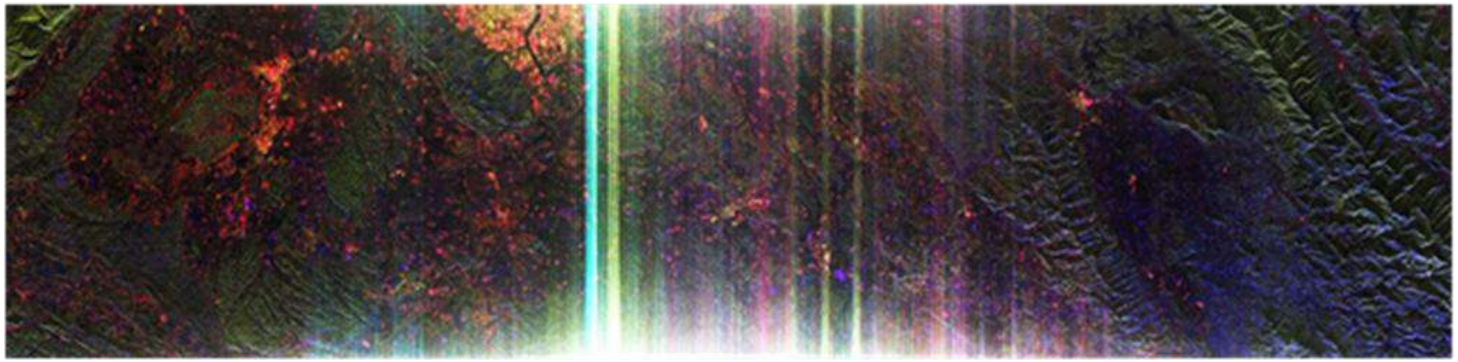
Folder / file	Description
Folder name	Start/stop dates and acquisition identifiers
annotation	• **navigation:** attitude and orbit XML files • **annot.xml:** raster info (axis, sampling), processing annotations. • **lut.nc**: 2D NetCDF ancillary layers (slopes, incidence and elevation angles, σ0, γ0, lat/lon/height, Faraday rotation, ionospheric phase screen)
measurement	• **.vrt file:** GDAL Virtual Format describing how to convert amplitude/phase to real/imaginary • GeoTIFF, multi-channel (HH-HV-VH-VV) • **abs.tiff:** amplitude ($\sqrt{\beta_0}$, absolute calibration) • **phase.tiff:** phase
preview	PNG preview images, kml
schema	XSD schemas for annot.xml and MPH xml
MPH xml	Main Product Header (MPH)

L1a processing: RFI mitigation

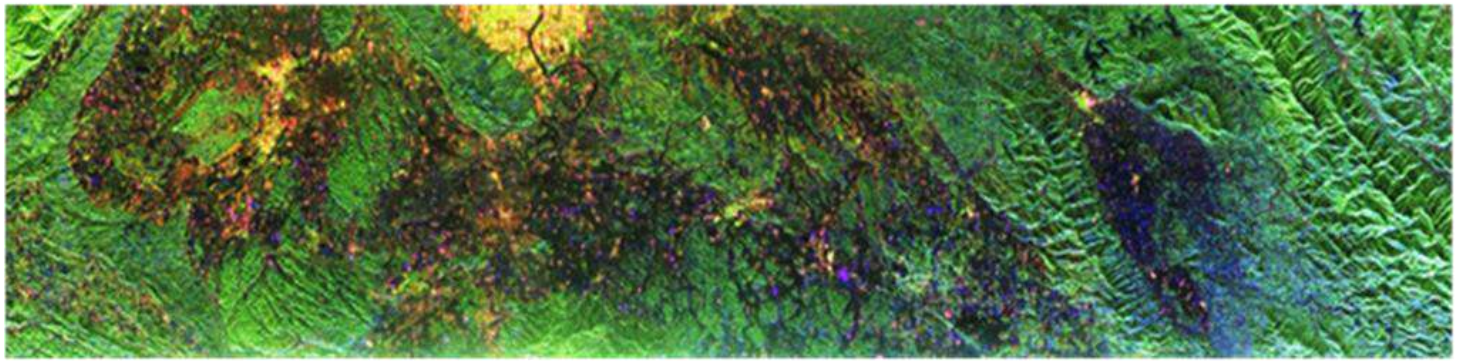
- Applied on raw data before other processing steps
- **Detection and mitigation of narrowband RFI**
  - comparison with reference chirp
  - notching affected frequencies



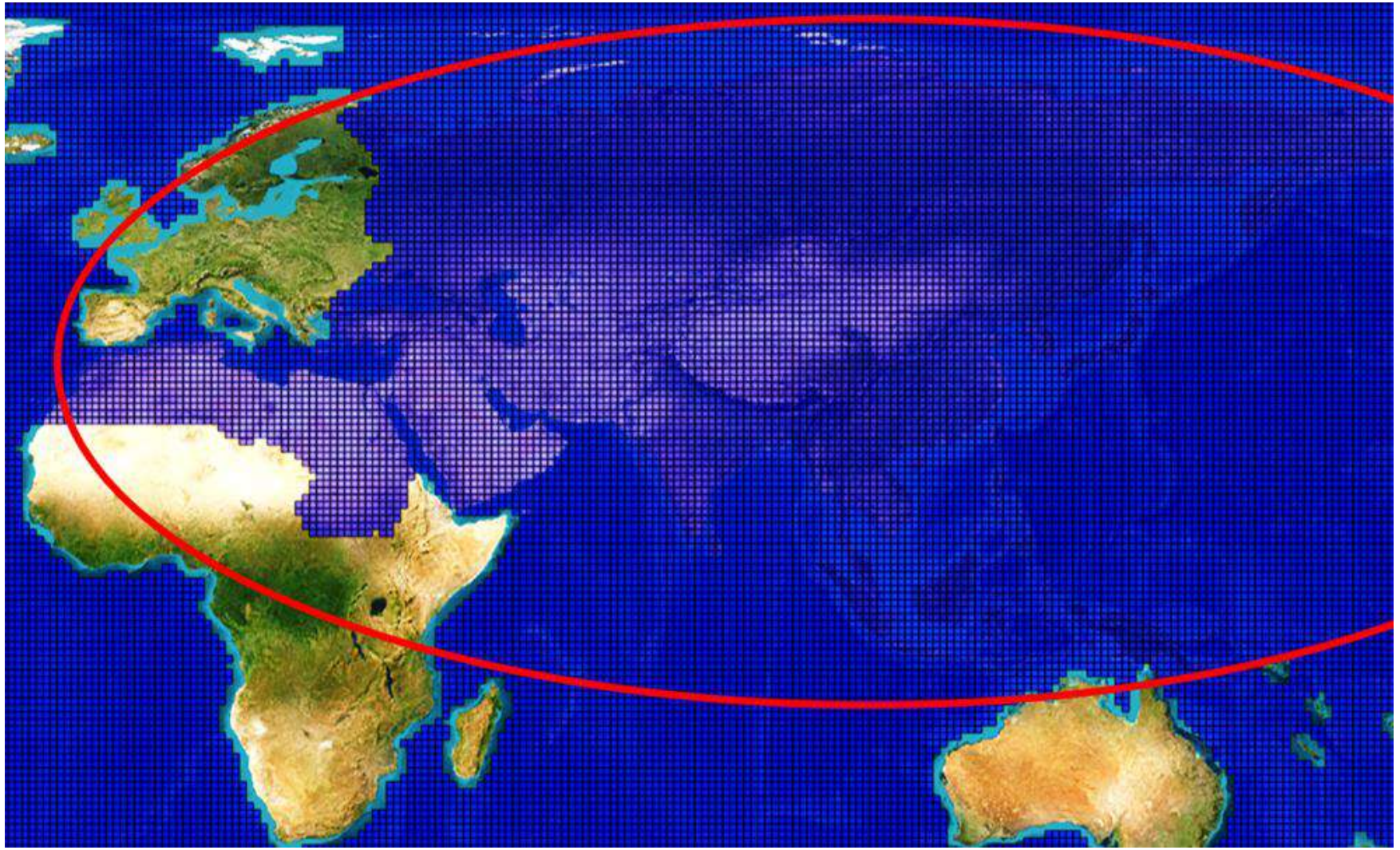
Mitigation OFF



Mitigation ON

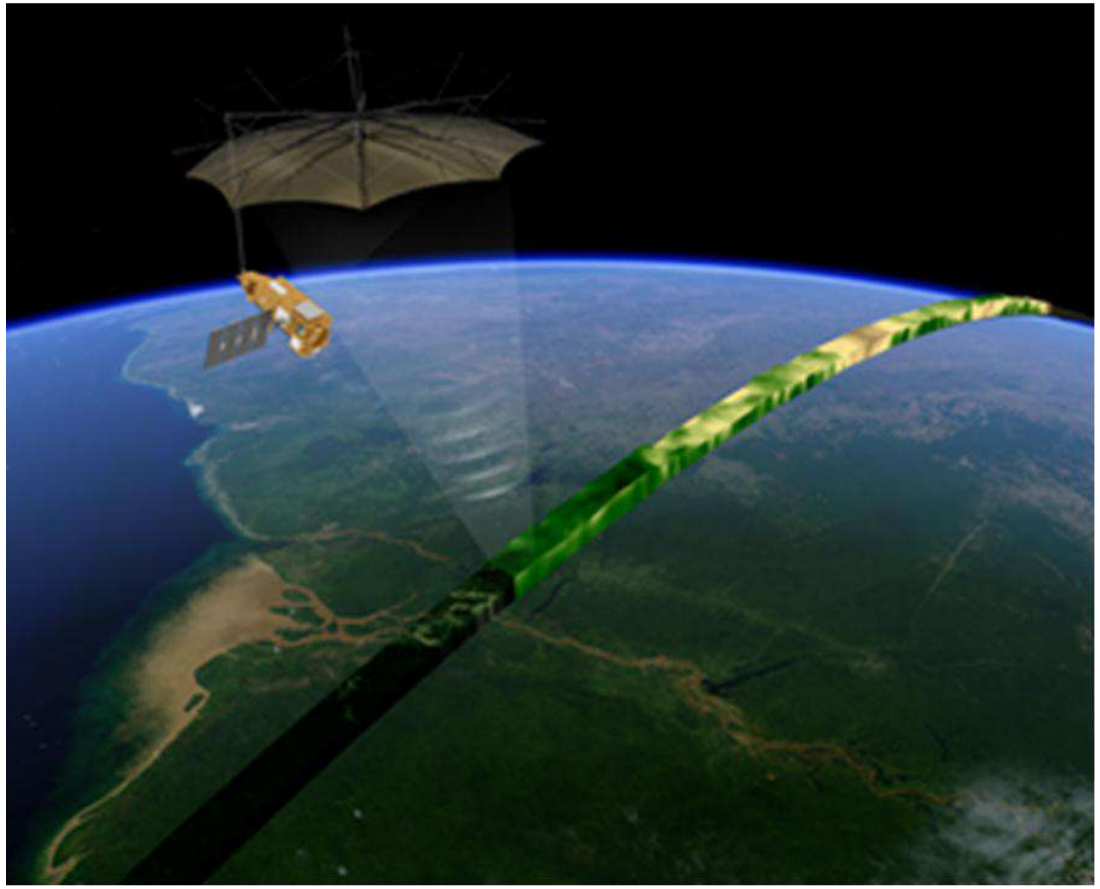


Where RFI mitigation is applied

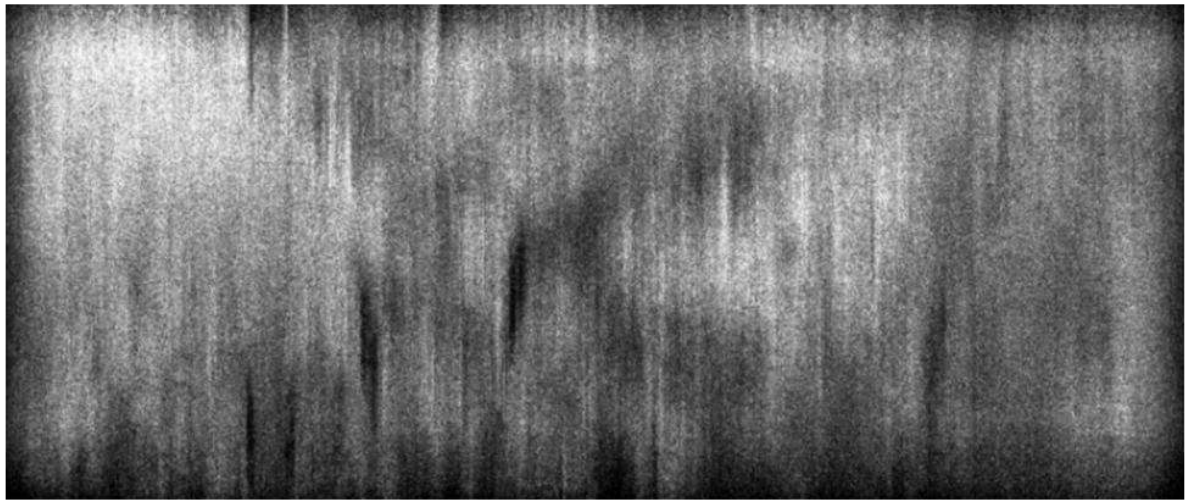


You may find something outside of this region

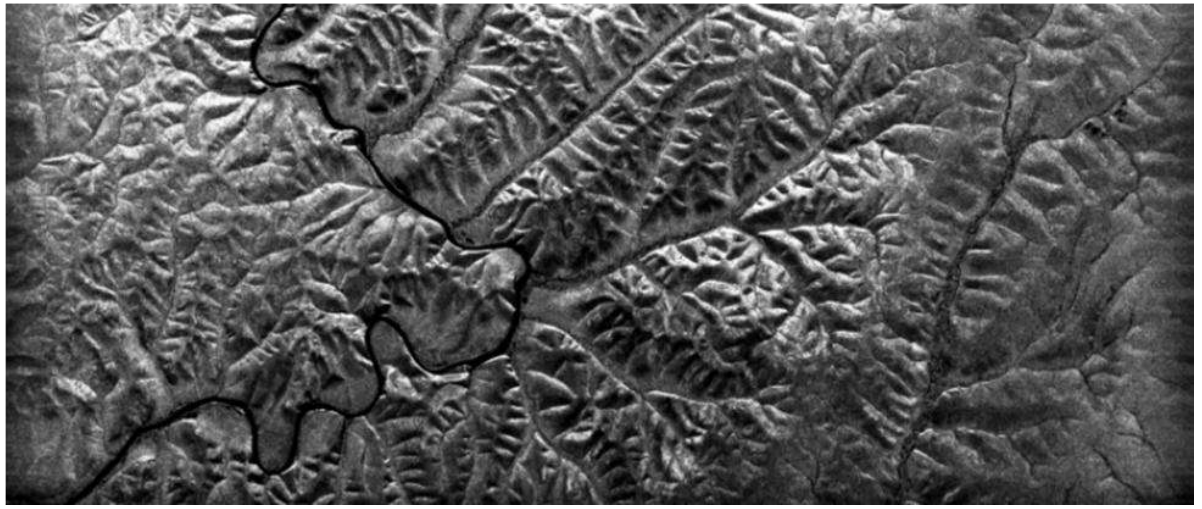
L1a processing: focusing



Raw data

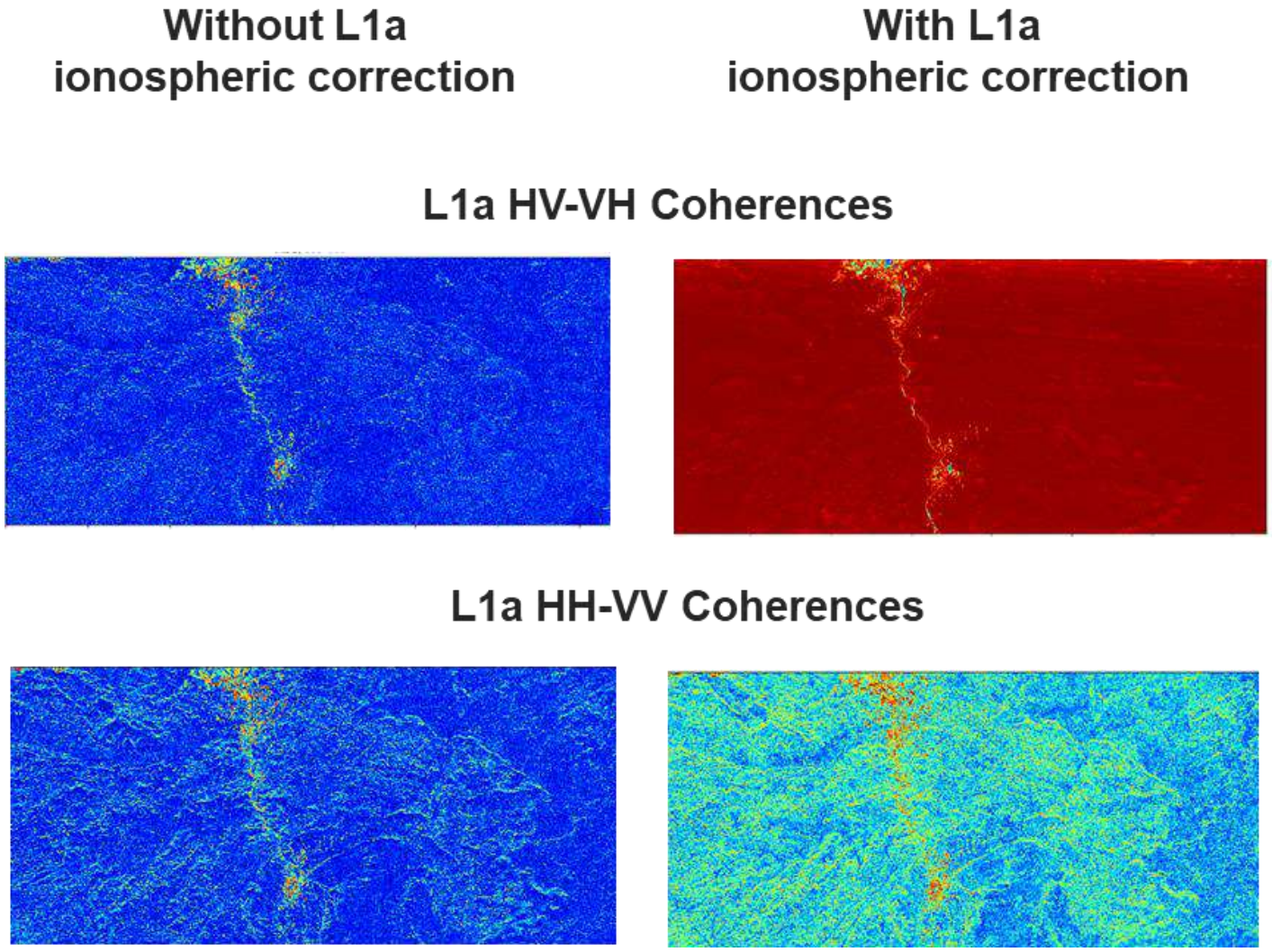


Focused data



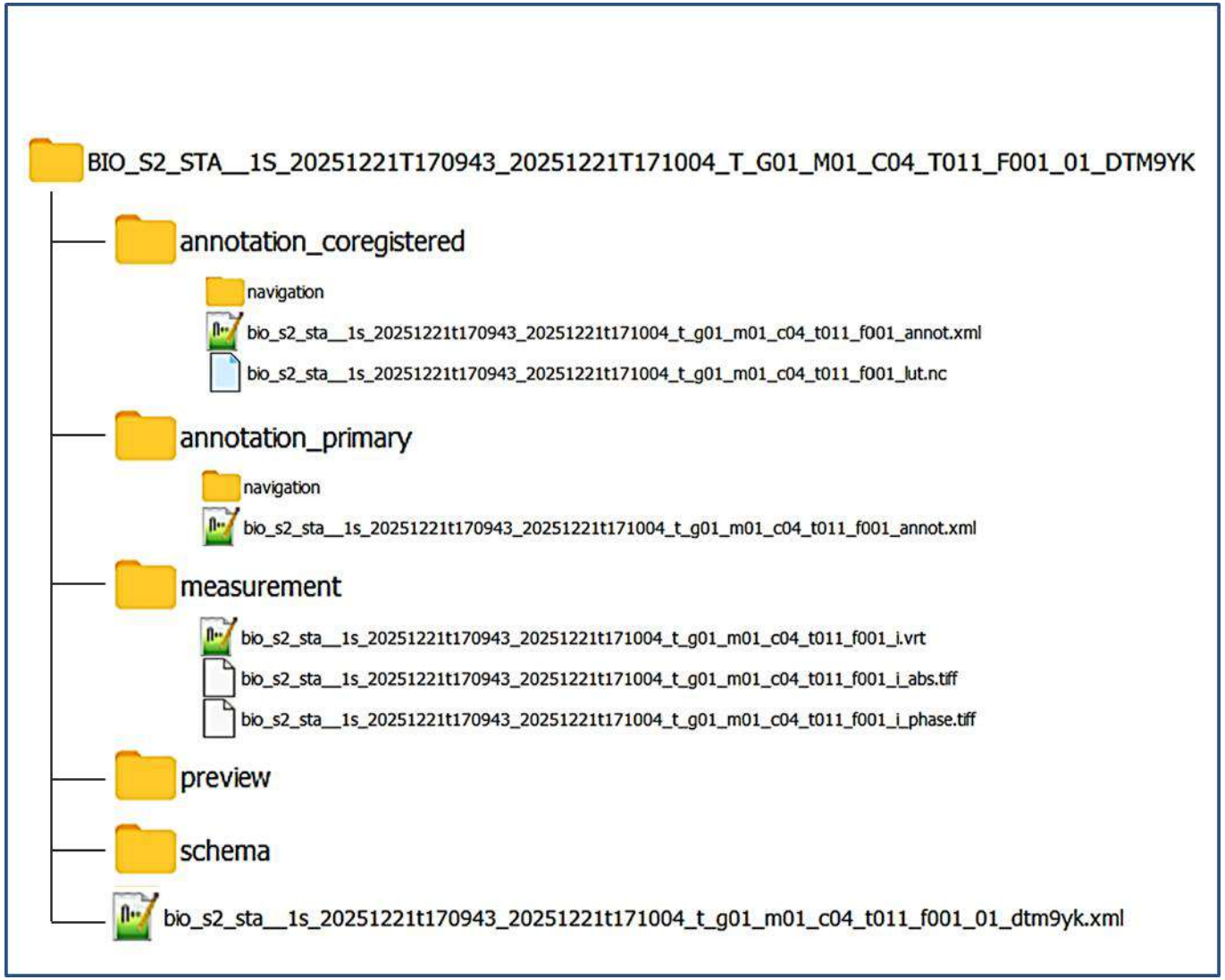
L1a processing: ionosphere

- Ionosphere effects on SAR
  - phase advance
  - group delay, meaning geolocation errors
  - Faraday rotation (FR), i.e. unbalanced polarimetry
- **Correction by estimation of FR on PolSAR**
- *Removes bulk effects*



Level-1c Product Format

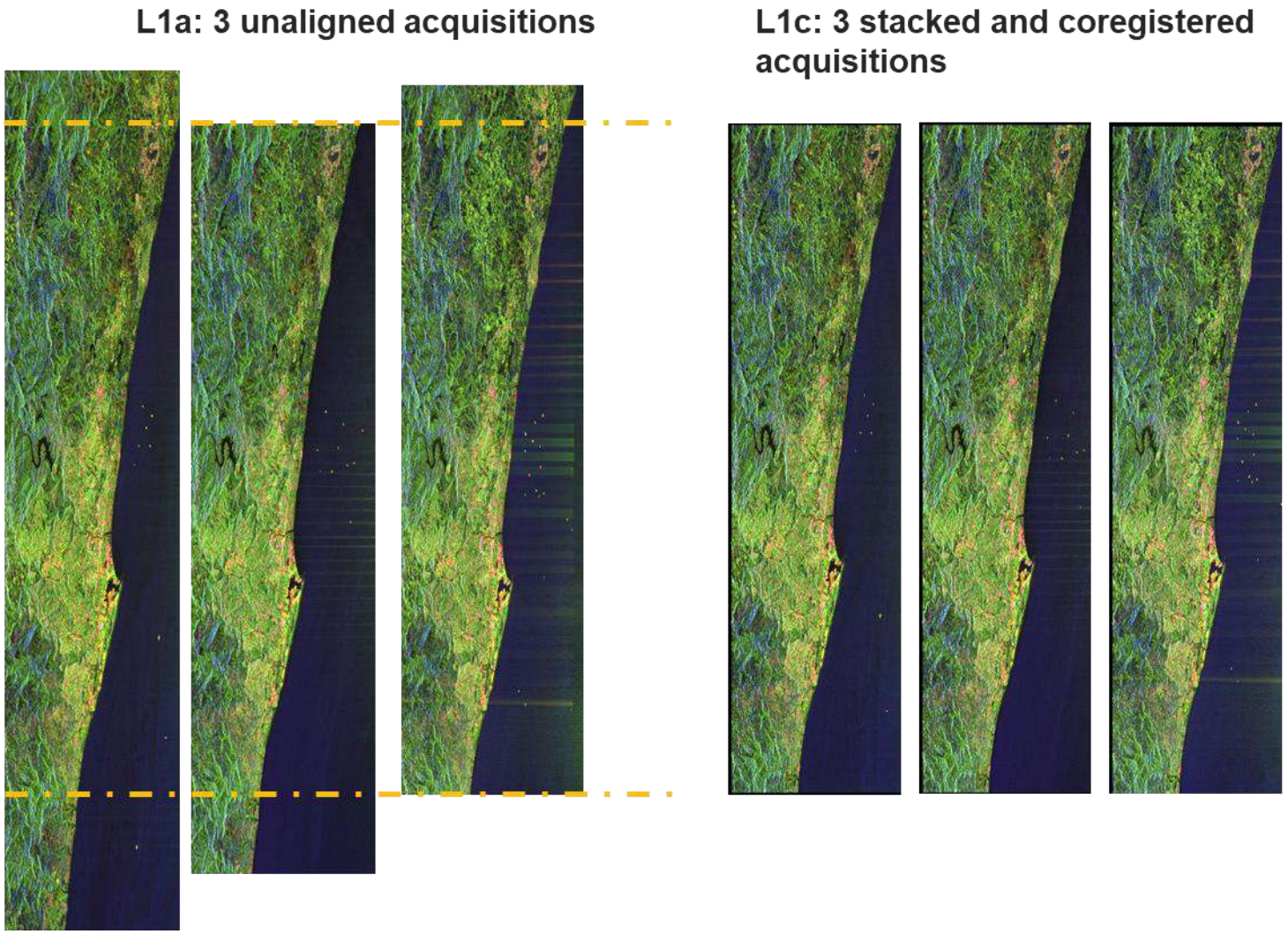
One folder per image in the stack



Folder / file	Description
Folder name	Start/stop dates and acquisition identifiers
annotation_ coregistered	• **navigation:** attitude and orbit XML files • **annot.xml:** raster info (axis, sampling), processing annotations. • **lut.nc**: 2D NetCDF ancillary layers (slopes, incidence and elevation angles, lat/lon/height, coregistration shifts, wavenumbers, flattening and SKP phases, coherence)
annotation_ primary	Annotations of the primary image used in coregistration
measurement	• **.vrt file:** GDAL Virtual Format describing how to convert amplitude/phase to real/imaginary • GeoTIFF, multi-channel as L1a: HH-HV-VH-VV or HH-XX-VV (XX = merged cross-pol)
preview	PNG preview images
schema	XSD schemas for annot.xml and MPH xml
MPH xml	Main Product Header (MPH)

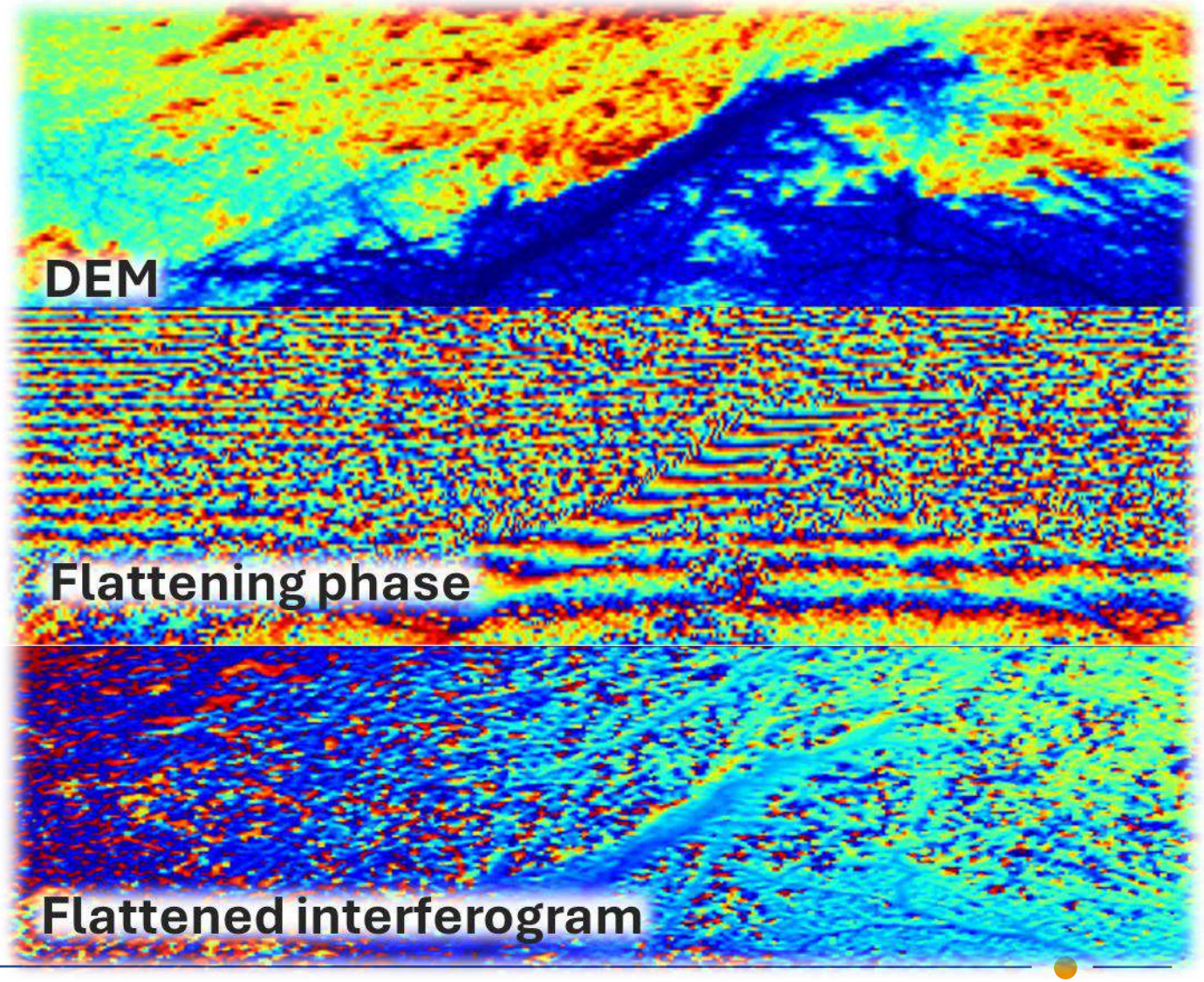
L1c processing: coregistration

- Acquisitions from different views must be stacked onto the same grid for InSAR processing
- **Coregistration**:
  1. nominal geometry and orbits
  2. incoherent speckle tracking
  3. resampling
- **Corrects (moderate) residual ionosphere**



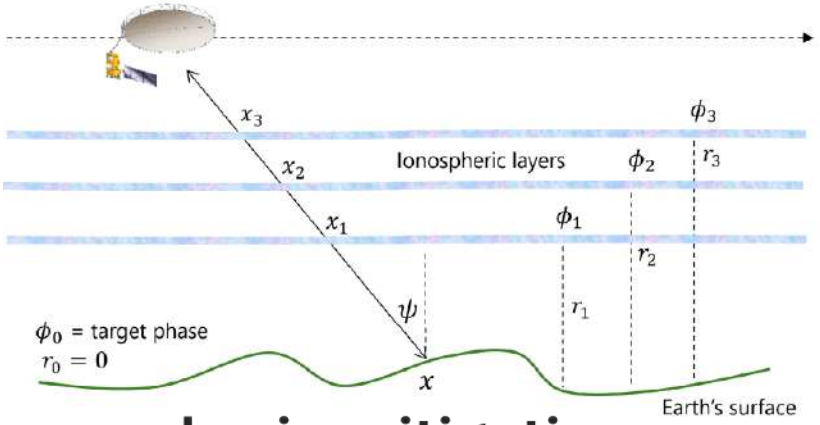
L1c processing: phase flattening

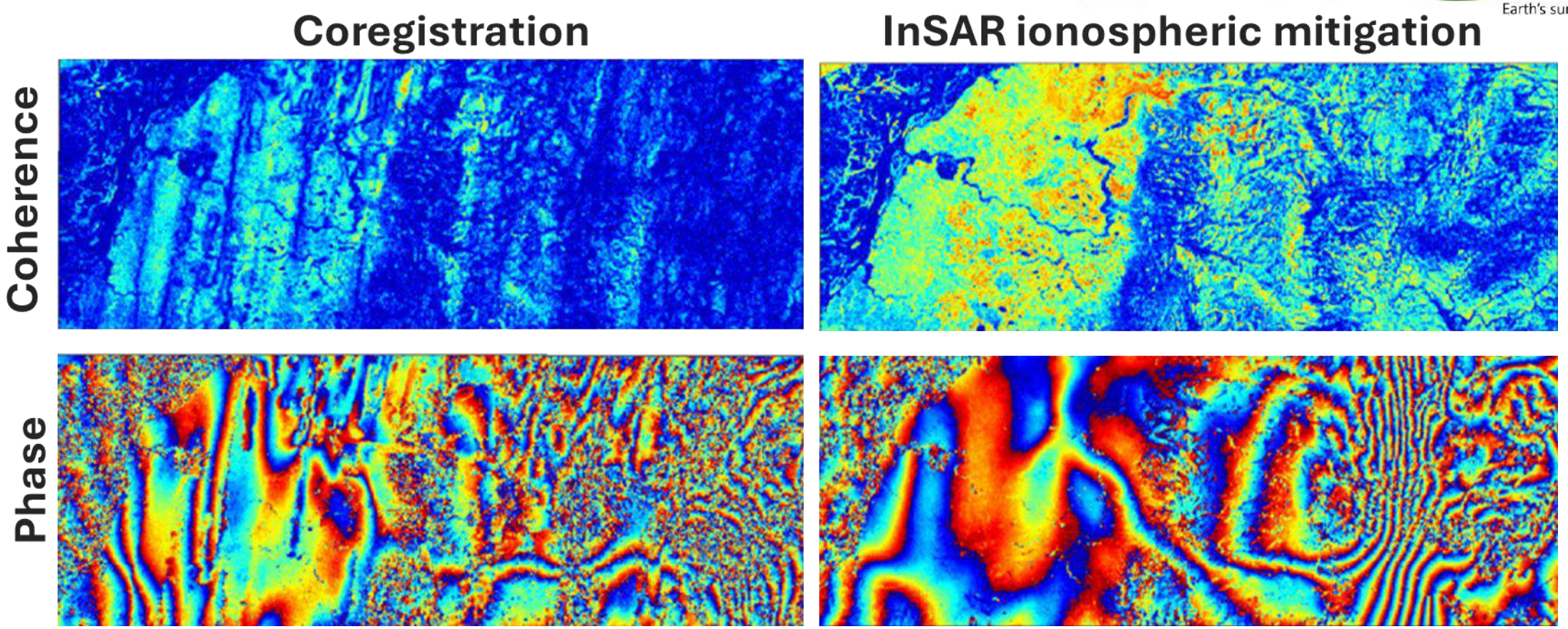
- **Nominal geometry is compensated** before InSAR processing
  - ➢ higher coherence
- InSAR phase contains
  - ○ residual topography/structure
  - ○ atmosphere



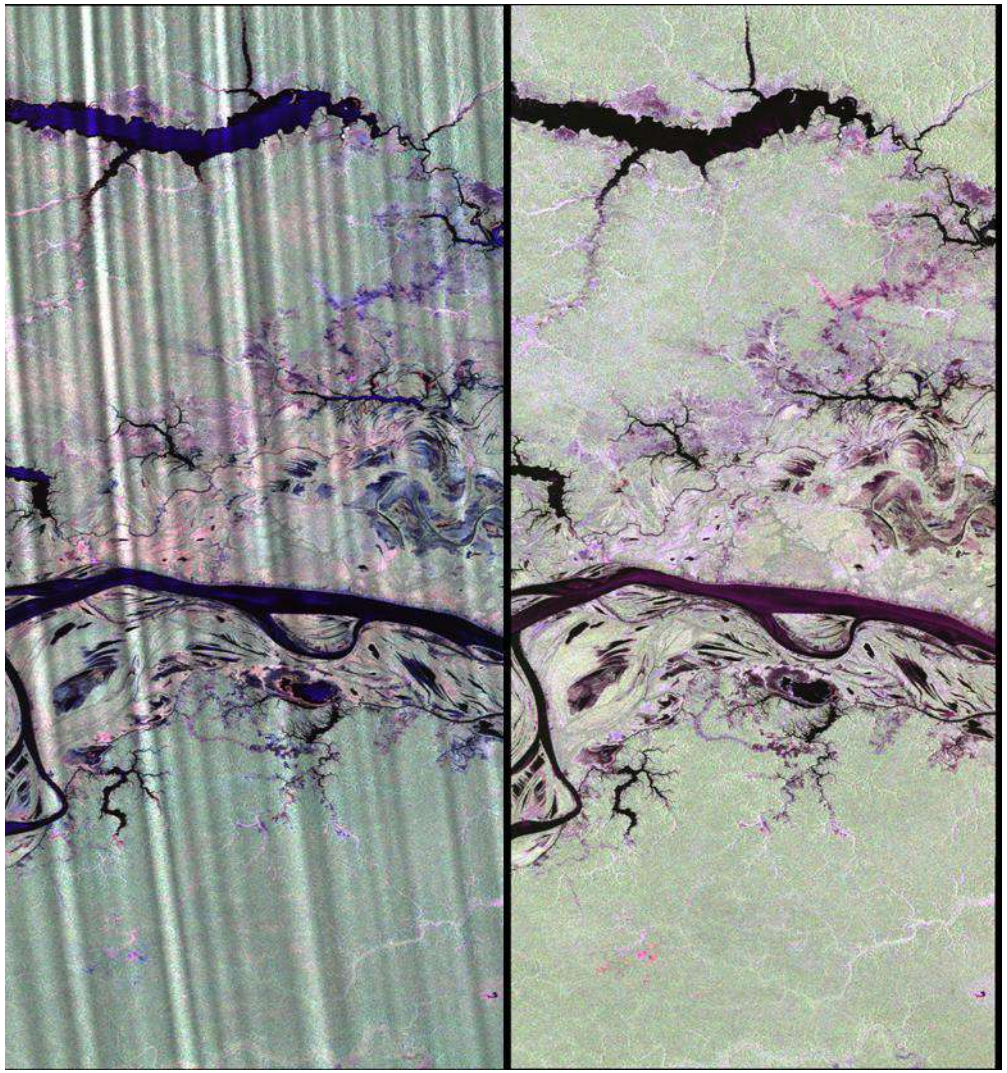
L1c processing: ionosphere

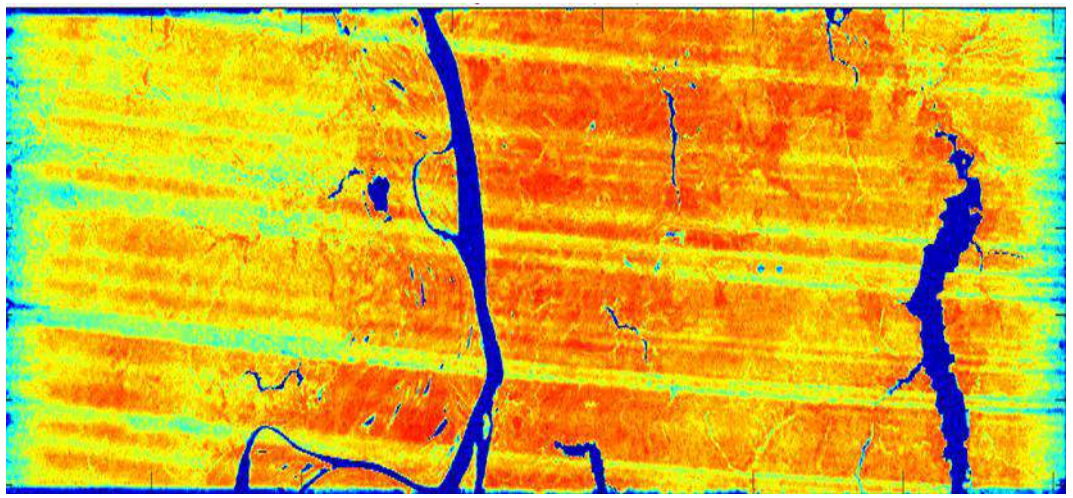
When ionosphere is severe, adhoc InSAR correction is applied leveraging azimuth diversity





Work in progress





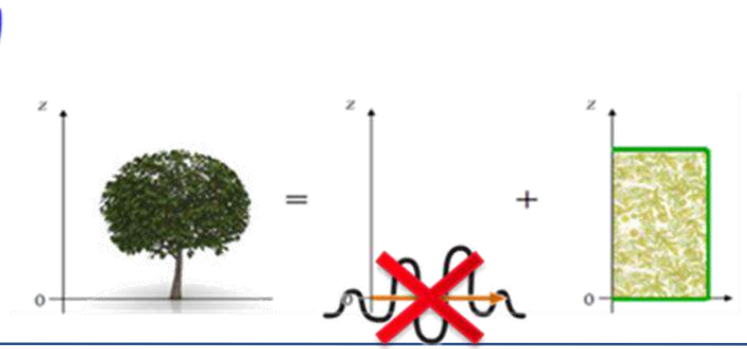
Sometimes fast varying strong ionosphere produces fluctuations and coherence loss

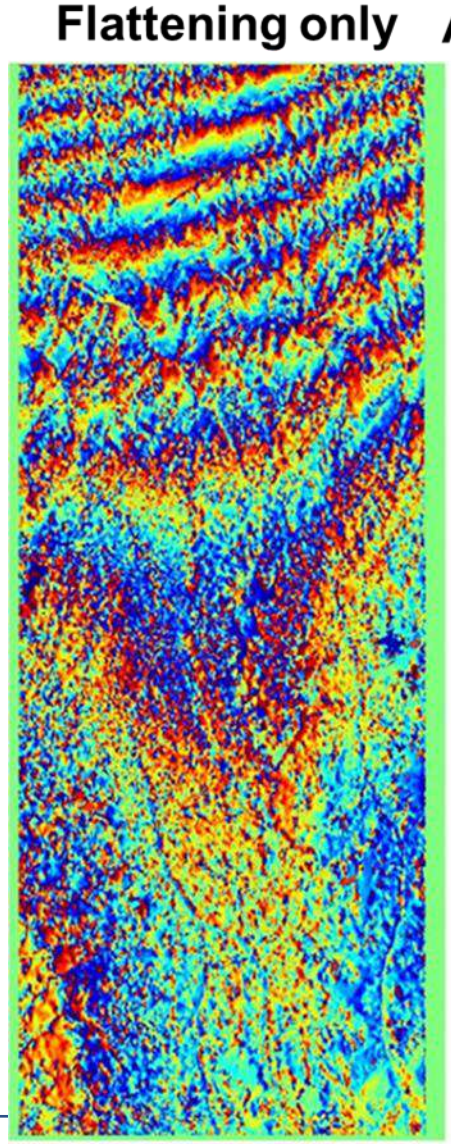
Level 1c processing: ground steering

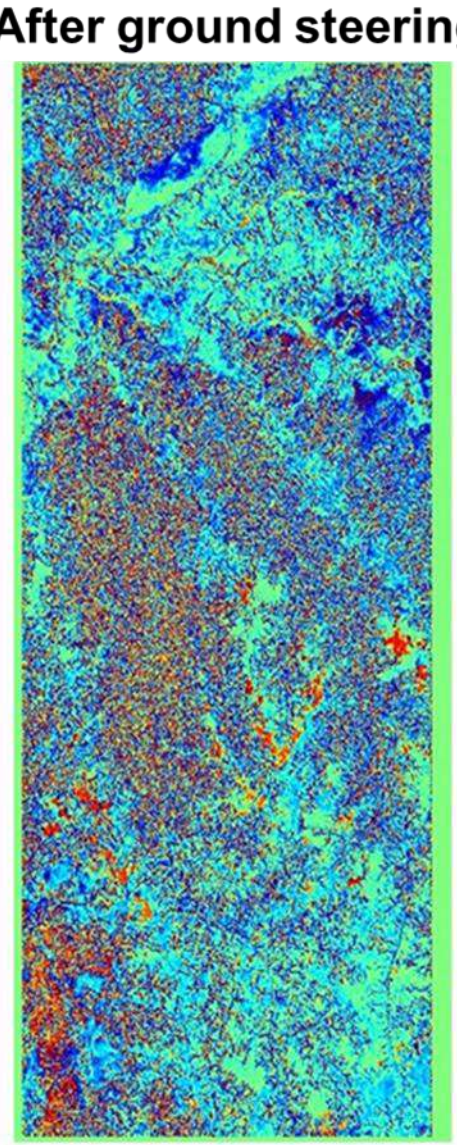
Phases estimated with InSAR or PolInSAR techniques

- **set ground level to zero height**
- **remove residual phase disturbances**





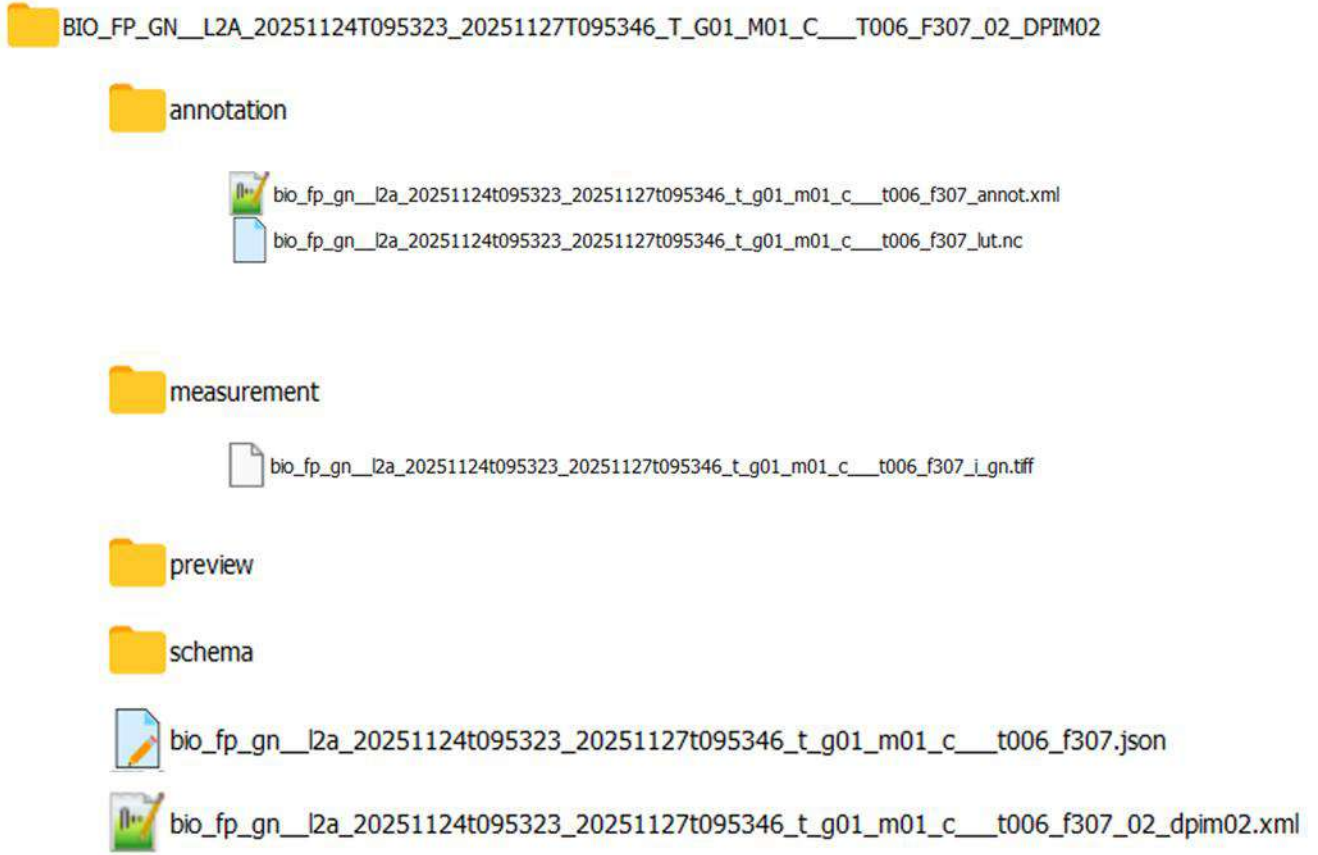




BIOMASS Level-2 Products

Level 2A (FH)	Level 2A (GN)
Forest Height	**Ground Notch**
Obtained by processing an entire L1C stack (TOM or INT).	Obtained by processing an entire L1C stack (TOM or INT).
Same L1 SCS coverage geocoded onto a Discrete Global Grid for forest products, **not tiled.**	Same L1 SCS coverage geocoded onto a Discrete Global Grid for forest products, **not tiled.**
Resolution: 200 m	Resolution: 100 m
Spacing: 3” x 3” to 3” x 6”	Spacing: 1.5” x 1.5” to 1.5” x 3”
Looks 4 x 25	Looks 2 x 12

Level-2a GN Product Format



Folder / file	Description
Folder name	Start/stop dates and acquisition identifiers
annotation	• **annot.xml:** raster info (axis, sampling), processing annotations, input (L1c) information • **lut.nc**: 2D NetCDF ancillary layers (local incidence angle, Forest Non-Forest mask: external mask, resampled on the product grid)
measurement	• GeoTIFF, ground-notched backscatter, multi-channel HH-XX-VV polarizations (XX = merged cross-pol)
preview	PNG preview images
schema	XSD schemas for annot.xml and MPH xml
JSON	JSON file for indexing products in SAR image databases
MPH xml	Main Product Header (MPH)

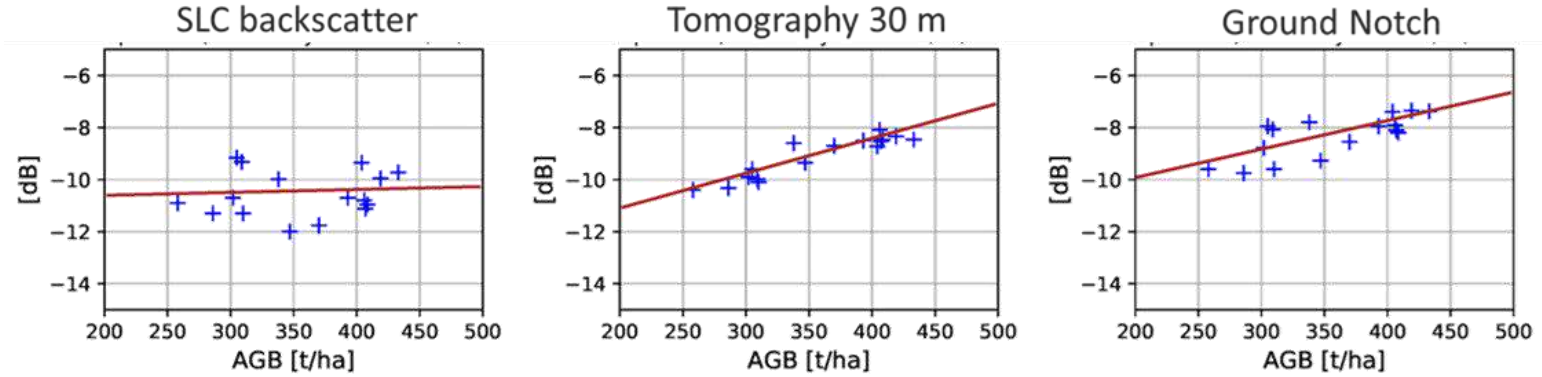
Level-2a FH Product Format

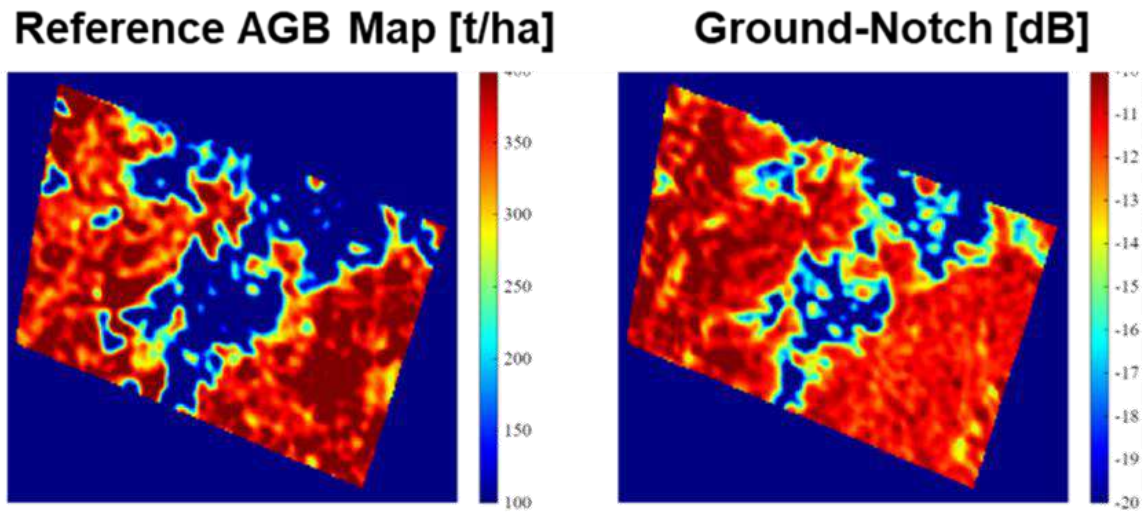


Folder / file	Description
Folder name	Start/stop dates and acquisition identifiers
annotation	• **annot.xml:** raster info (axis, sampling), processing annotations, input (L1c) information • **lut.nc**: 2D NetCDF ancillary layer (Forest Non-Forest mask: external mask, resampled on the product grid)
measurement	• GeoTIFF • **fh.tiff:** Top Canopy Height [m] • **quality.tiff:** Quality layer (inversion performance index)
preview	PNG preview images
schema	XSD schemas for annot.xml and MPH xml
JSON	JSON file for indexing products in SAR image databases
MPH xml	Main Product Header (MPH)

Level 2a processing: ground notch

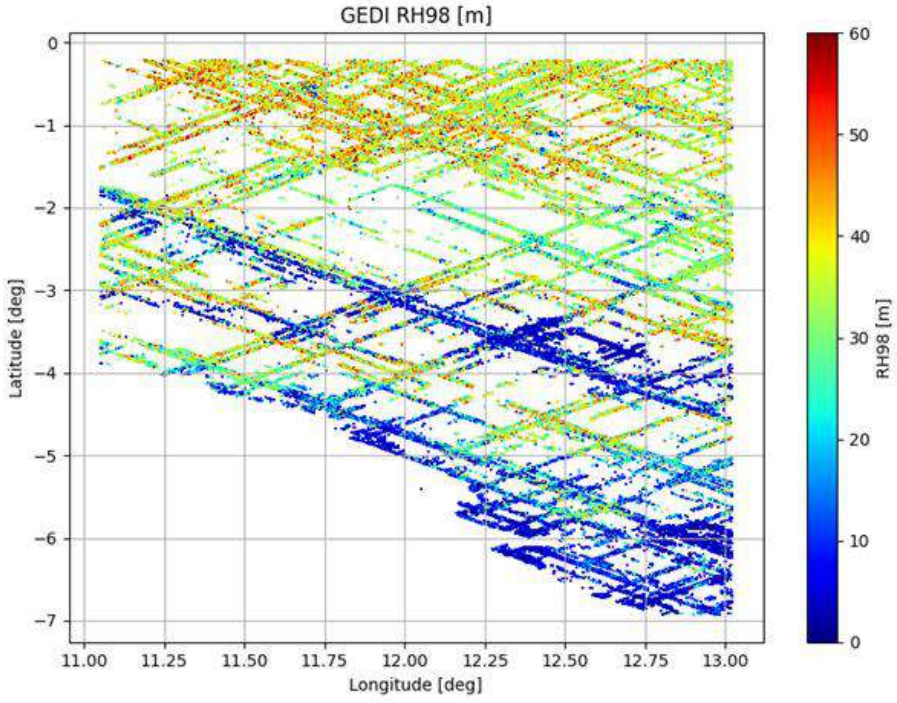
- Ground notch (GN) is **InSAR filtering to isolate forest backscatter**
  - Tomography with full stack gives 3D ⇒ take upper layers
  - GN can work with as low as 2 acquisitions
- **Sensitivity to AGBD is improved** w.r.t. single acquisition

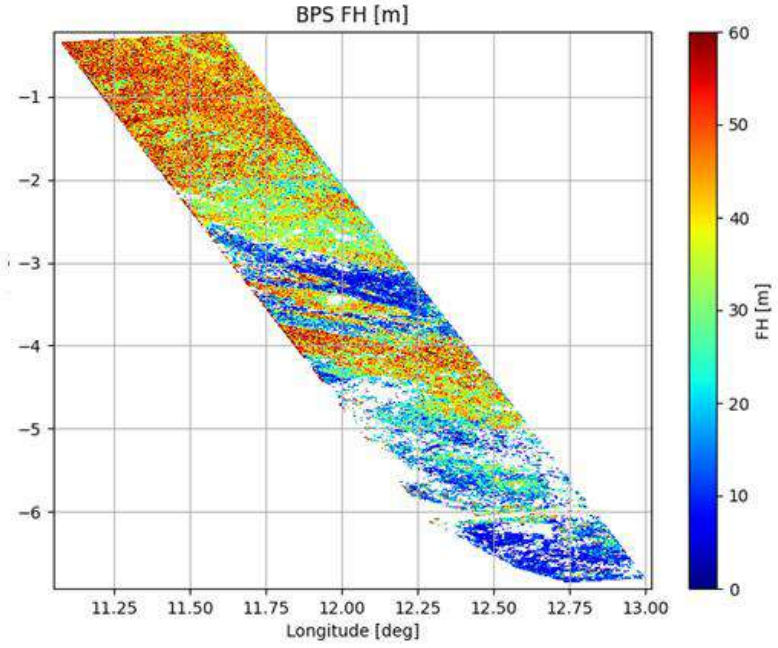


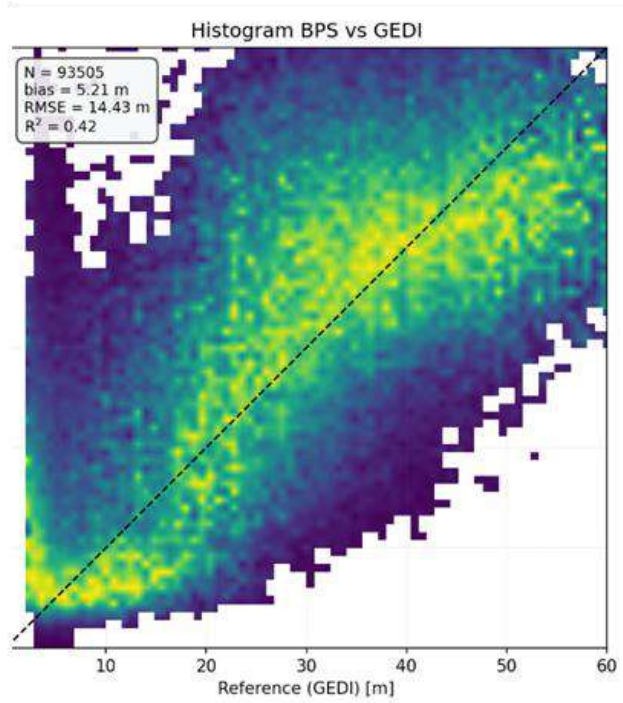


Level 2a processing: forest height

- A two-component model (ground and canopy) is fit to calibrated InSAR coherence
- Leveraging one or **two InSAR baselines for sensitivity to different heights**

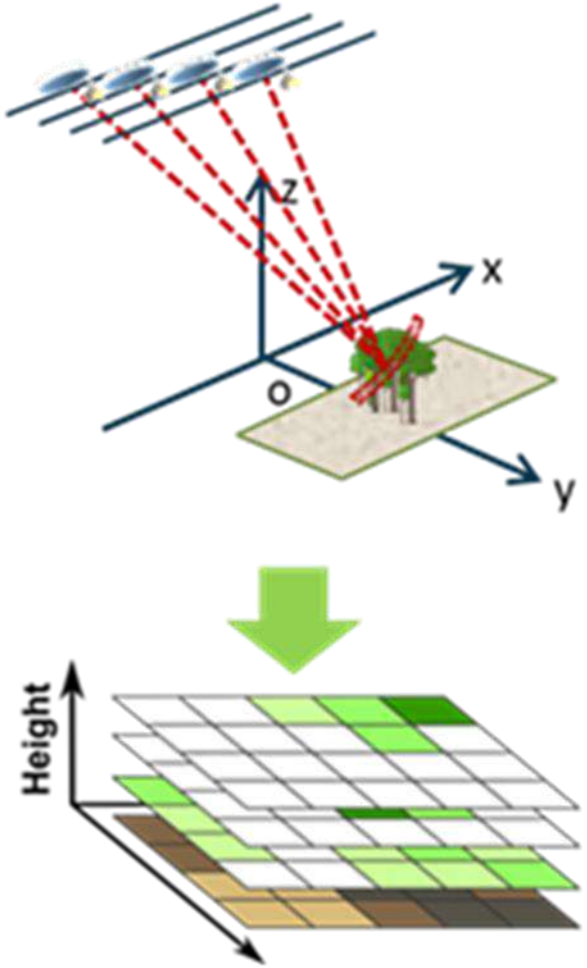






Experimental Products: Tomography

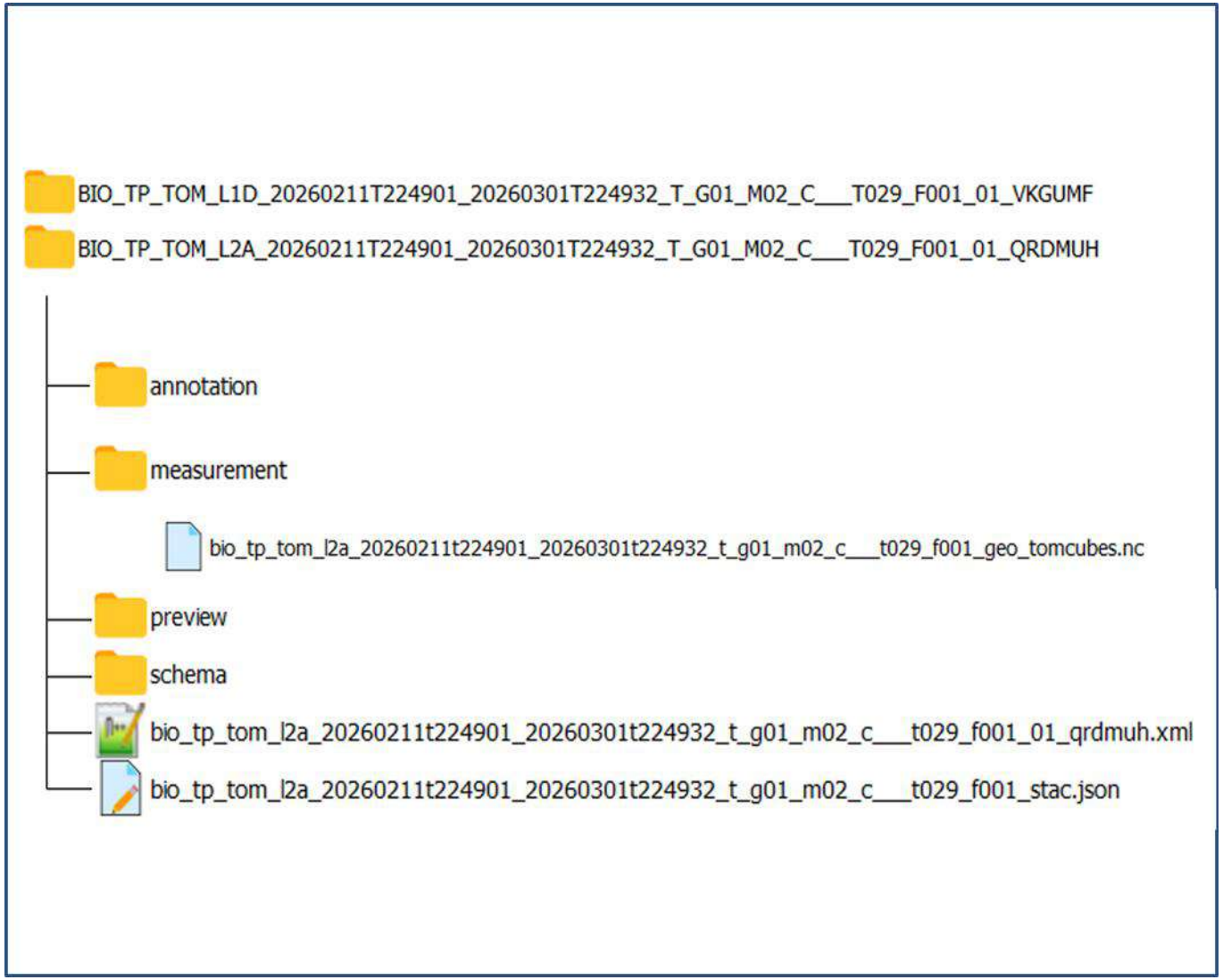
TomoSAR



$$\gamma_{nm} \cong \int f(z) \exp(jk_z^{nm} z) dz$$

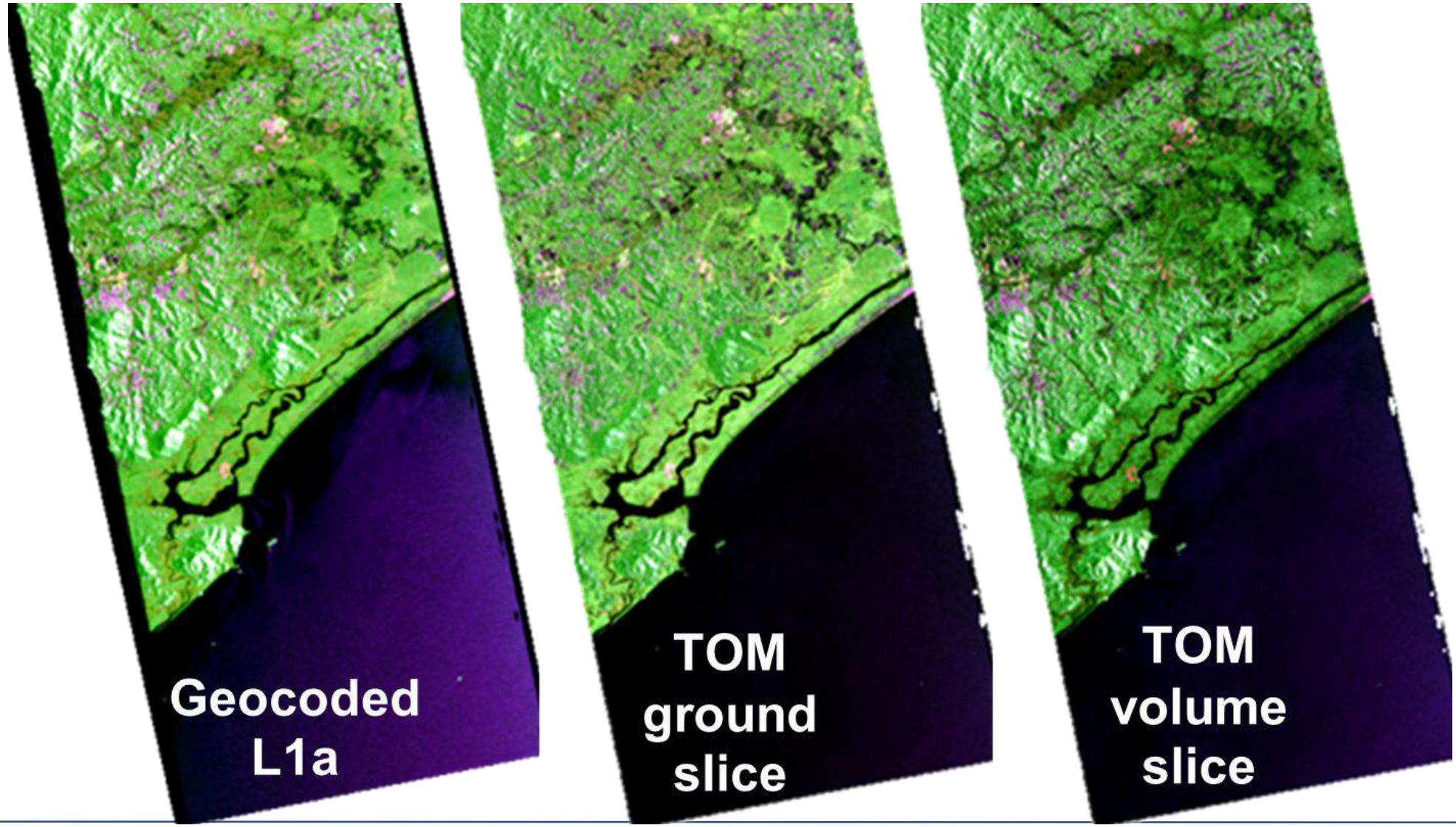
- BIOMASS acquires different views of the scene
  - **3D voxels can be reconstructed!**
- Not yet available on MAAP
- **Exclusively available to you for Hackaton!**

TOM Product Format

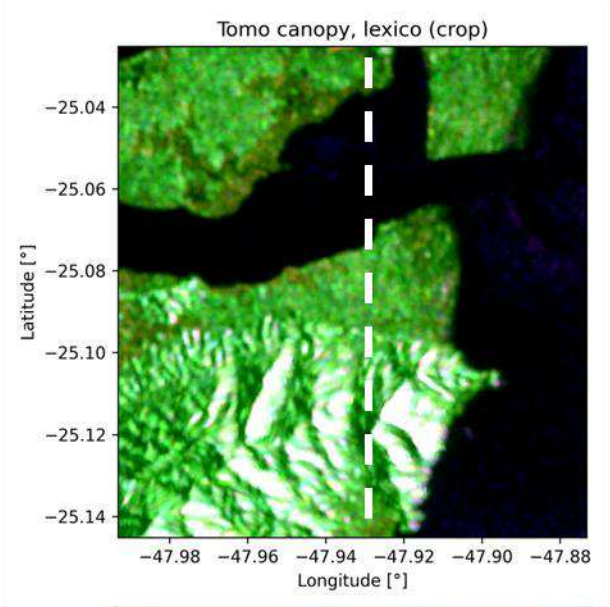


Folder / file	Description
Folder name	Start/stop dates and acquisition identifiers Two products: • L1D: slant-range geometry • L2A; geocoded geometry
annotation	Same files as in the L2a products: annot.xml and lut.nc (ancillary layers)
measurement	**geo_tomcubes.nc:** 3D NetCDF with the tomogram, one backscatter layer for each height (HH, HV, VV)
preview	PNG preview images
schema	XSD schemas for annot.xml and MPH xml
JSON	JSON file for indexing products in SAR image databases
MPH xml	Main Product Header (MPH)

TOM example: Ilha do Cardoso, Brazil

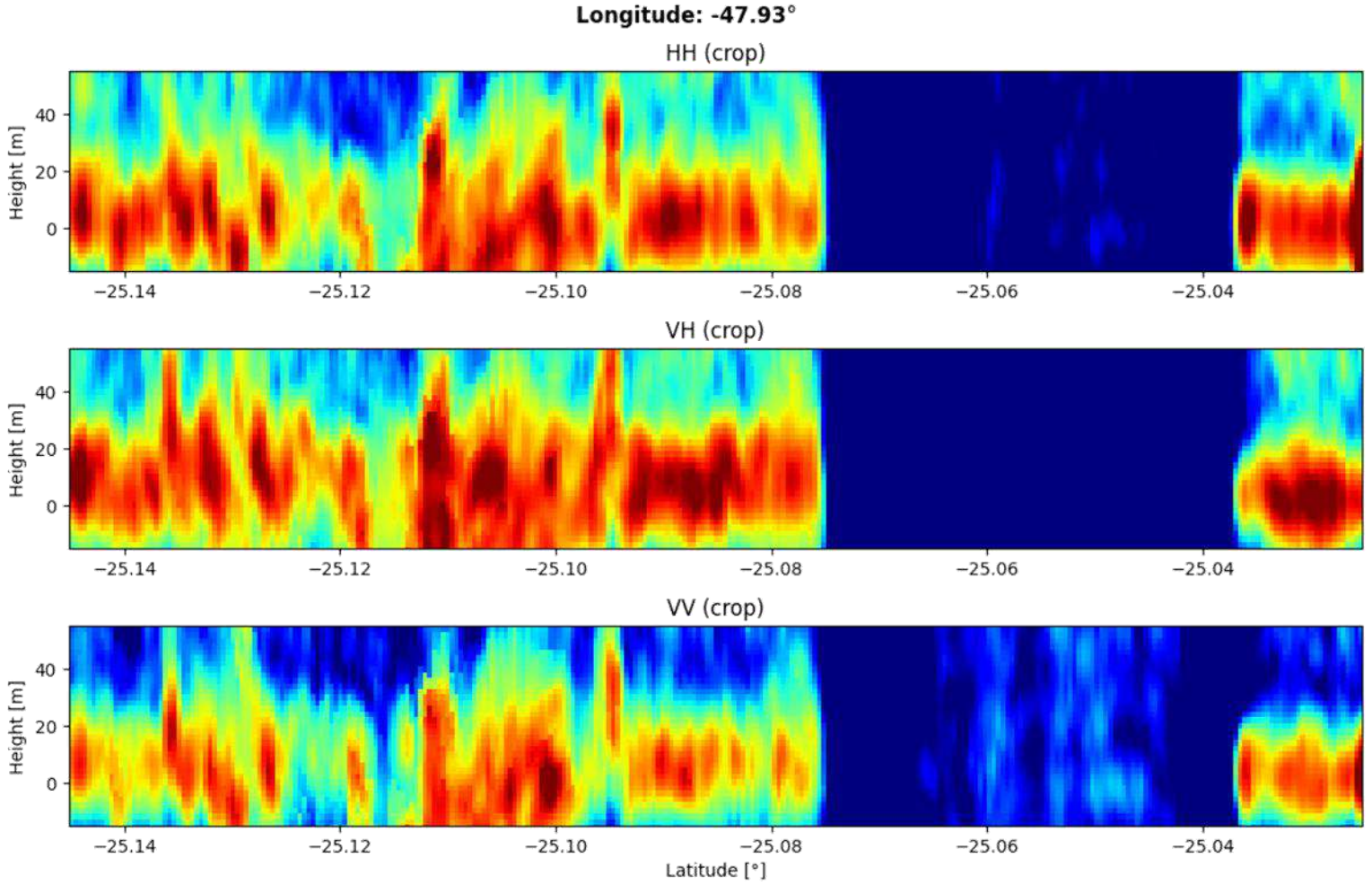


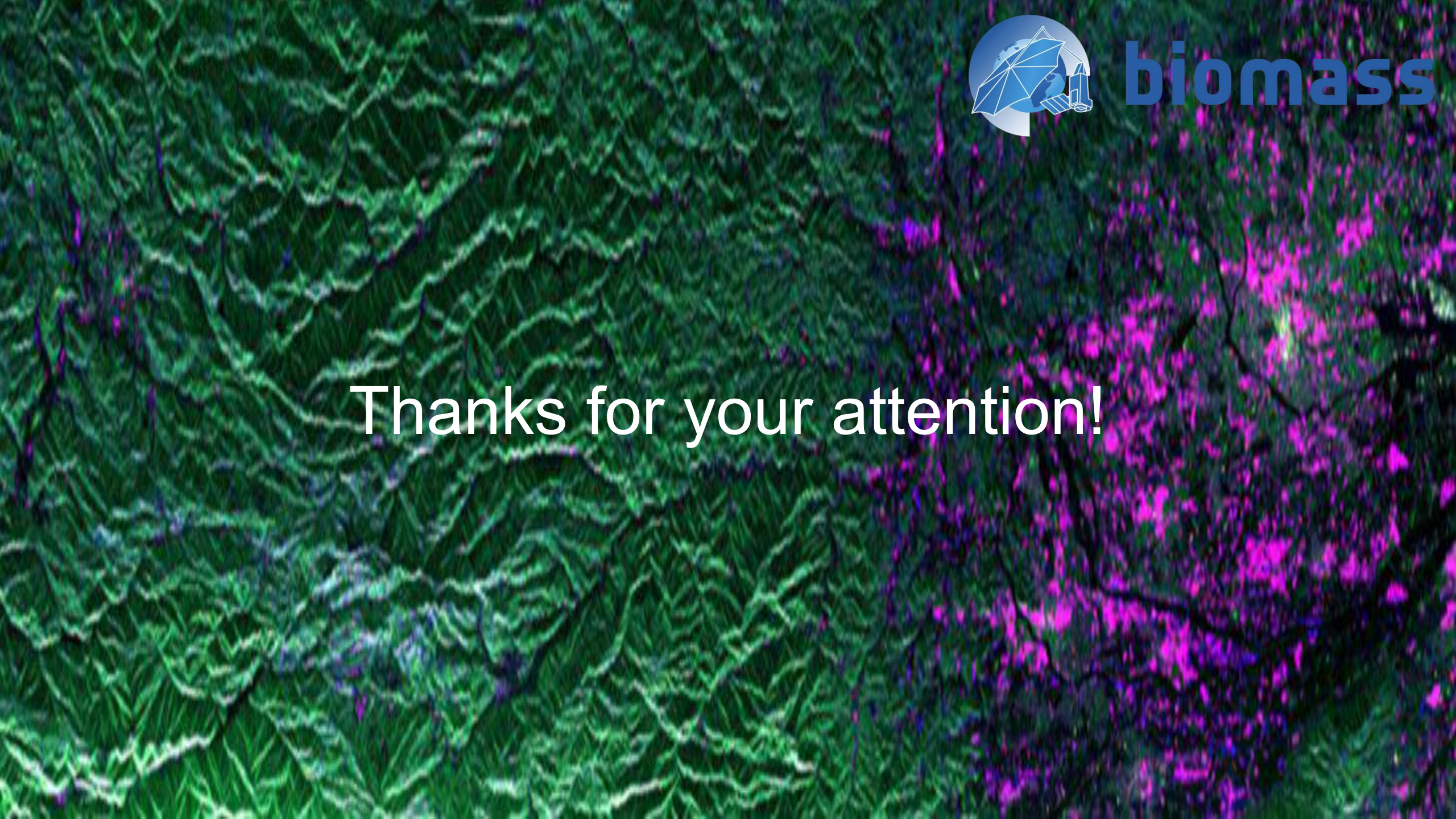
TOM example: Ilha do Cardoso, Brazil



Optical Image






biomass
Thanks for your attention!

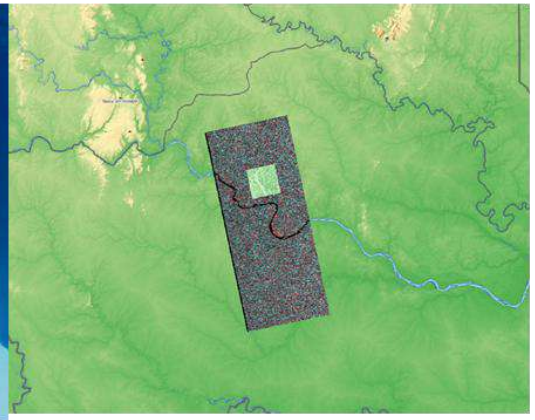
Data hints for the challenges

Thanks to Marco, Marco & Emanuele for searching the datasets!

Forests

Colombia: moderate slopes, dense uniform forest



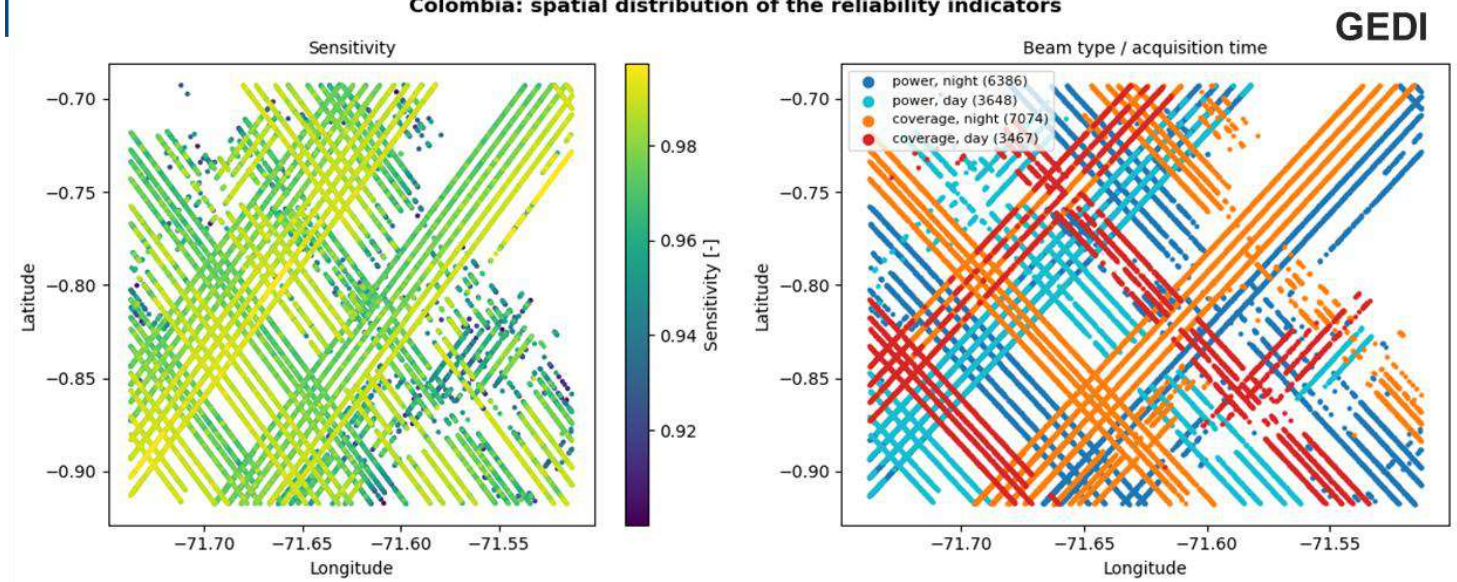


LAT_MIN = -0.88
LAT_MAX = -0.73
LON_MIN = -71.70
LON_MAX = -71.55

This patch features

- nearly flat terrain without promontories and small hills with moderate slopes.
- Based on satellite imagery, the forest type appears to be very uniform and of the 'dense rainforest' type

Colombia: spatial distribution of the reliability indicators



Colombia

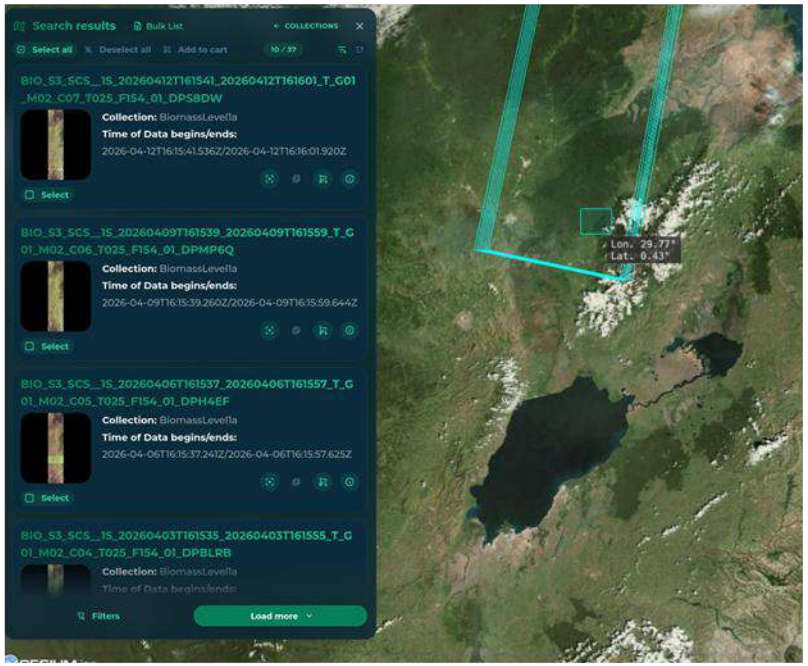
PRODUCT LEVEL	DATA TAKE ID
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L1A Preview



Congo : Stanley mountain zone, prominent slopes





Lat: 0.4 0.5
Lon: 29.7 29.8

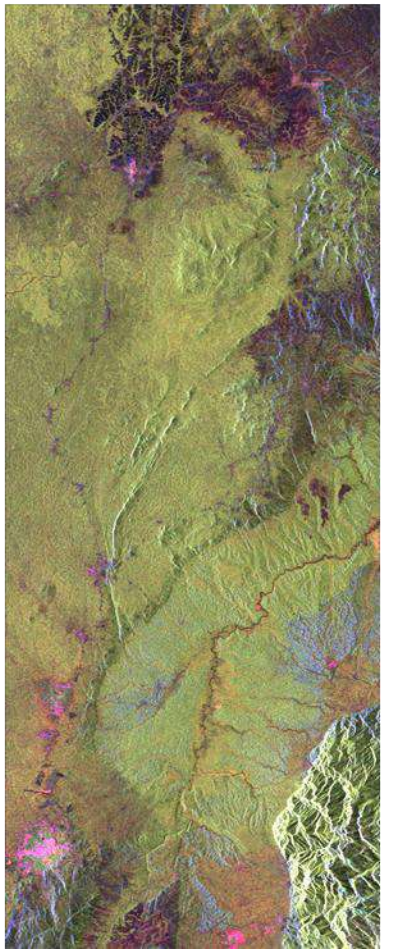
The region features:

- very dense vegetation and a very interesting topography:
  - within just a few kilometers, it transitions from the plain to a mountainous area with prominent relief.
  - Interesting: the mountain is completely covered in dense vegetation similar to that found in the plains."

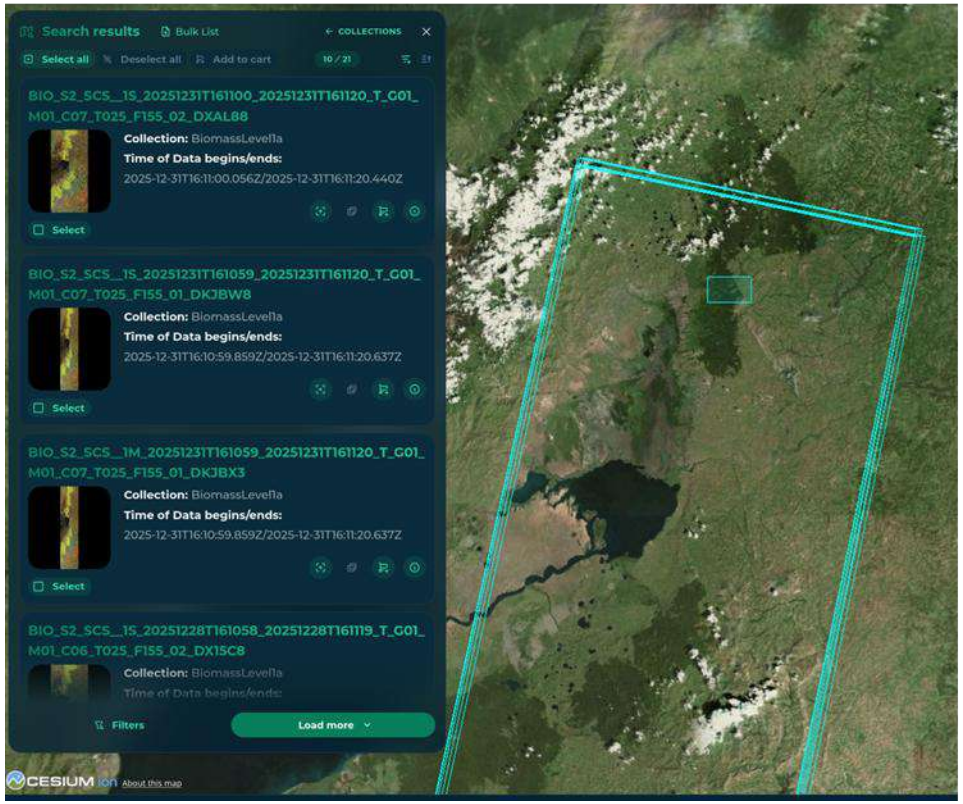
Congo

PRODUCT LEVEL	DATA TAKE ID
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	BIO_S3_SCS__1S_20260412T161541_20260412T161601_T_G01_M02_C07_T025_F154_01_DPS8DW

L1A Preview



Uganda : George Lake, forest/agricultural areas



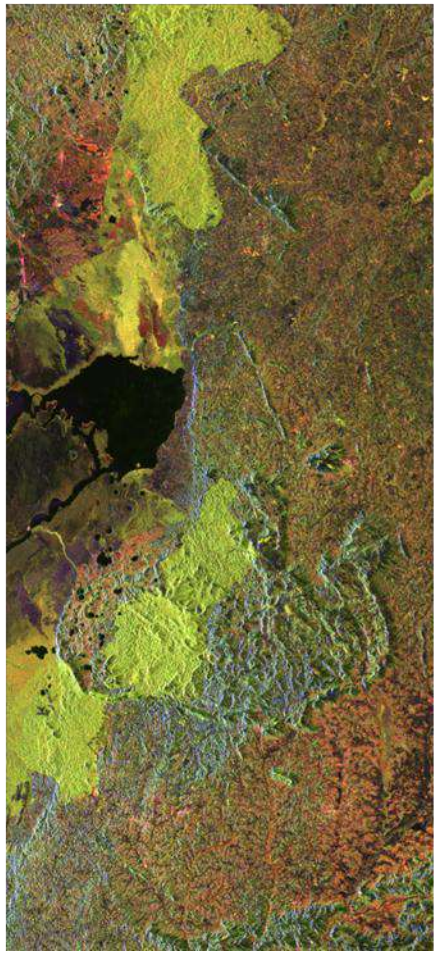
lat: 0.3 0.4
lon: 30.3 30.4

The region features:

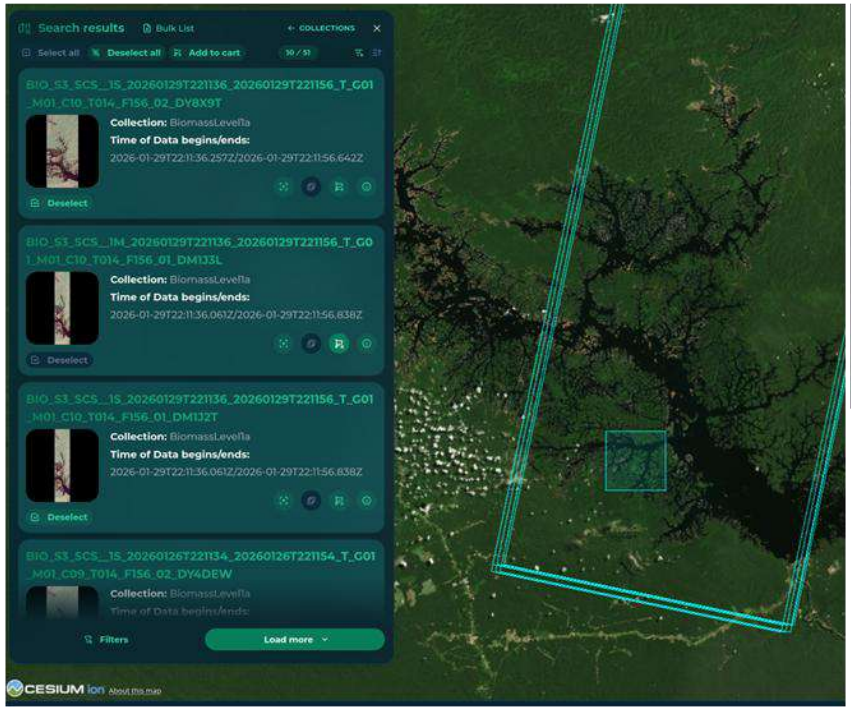
- relatively hilly terrain
- the forest exhibits a unique distribution: there is a clear separation between forested and agricultural areas, in some places even demarcated by a straight line. This could be interesting as it might simplify forest/non-forest discrimination work and enable targeted studies."

Uganda

PRODUCT LEVEL	DATA TAKE ID
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Brazil, Balbina hydroelectric basin





Environmental Context

Site of a major 1970s ecological disaster caused by hydropower plant construction in the Amazon rainforest.

Reservoir inundation submerged large forest areas, leaving dead trees standing *in situ*.

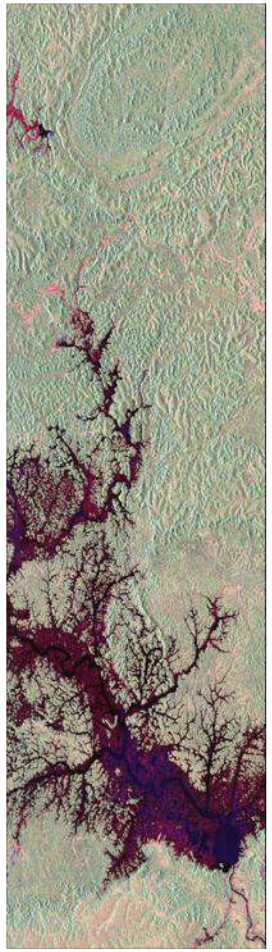
Site Characteristics

Shallow reservoir areas expose vertical standing trunks above the water surface.

Opportunity to study biomass signal behavior under flooded rainforest conditions.

Brazil, Balbina

L1A Preview

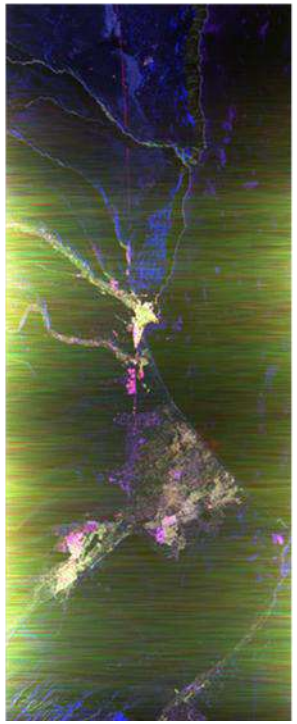


PRODUCT LEVEL	DATA TAKE ID
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	BIO_S3_SCS__1S_20260129T221136_20260129T221156_T_G01_M01_C10_T014_F156_01_DM1J2T
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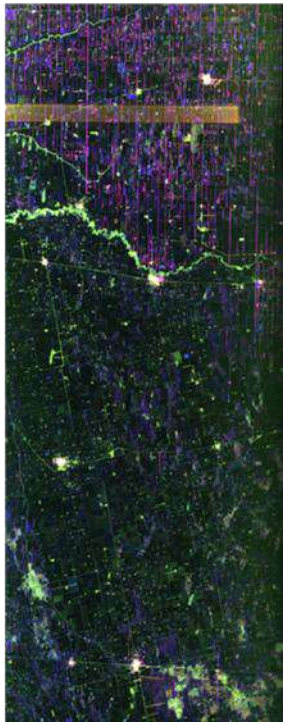
RFI

PRODUCT LEVEL	DATA TAKE ID	
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	BIO_S1_SCS__1S_20260918T104025_20260918T104046_T_G01_M05_C02_T021_F293_02_DXZAG1	BIO_S1_SCS__1S_20260919T220551_20260919T220611_T_G01_M05_C02_T043_F184_02_DY1SIO
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	BIO_S1_SCS__1S_20260924T104030_20260924T104050_T_G01_M05_C04_T021_F293_02_DYA6ZC	BIO_S1_SCS__1S_20260925T220555_20260925T220616_T_G01_M05_C04_T043_F184_02_DYCSA3
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L1A Prev #1

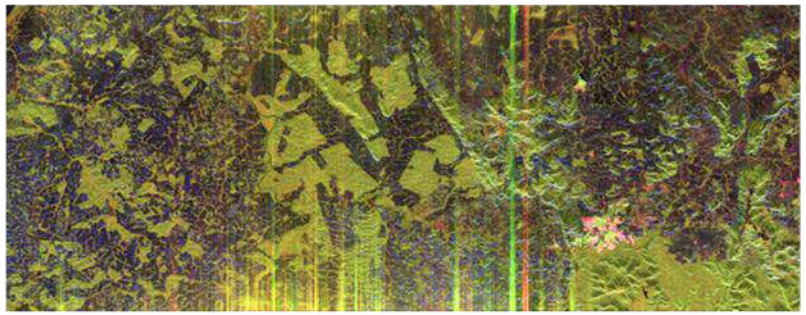


L1A Prev #2

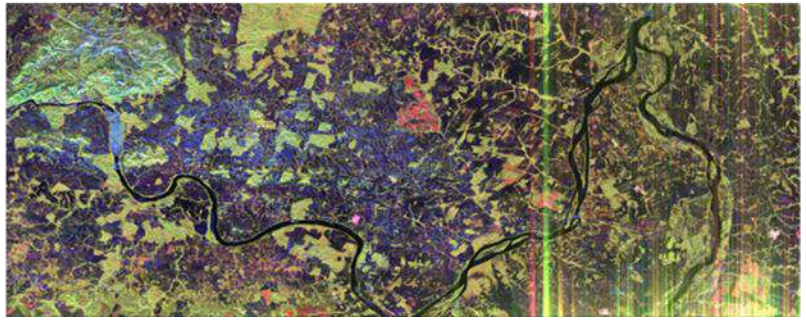


PRODUCT LEVEL	DATA TAKE ID	
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	BIO_S1_SCS__1S_20260918T212530_20260918T212551_T_G01_M05_C02_T028_F160_02_DY00H6	BIO_S1_SCS__1S_20260918T090622_20260918T090642_T_G01_M05_C02_T020_F306_02_DXZ169
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L1A Prev #1

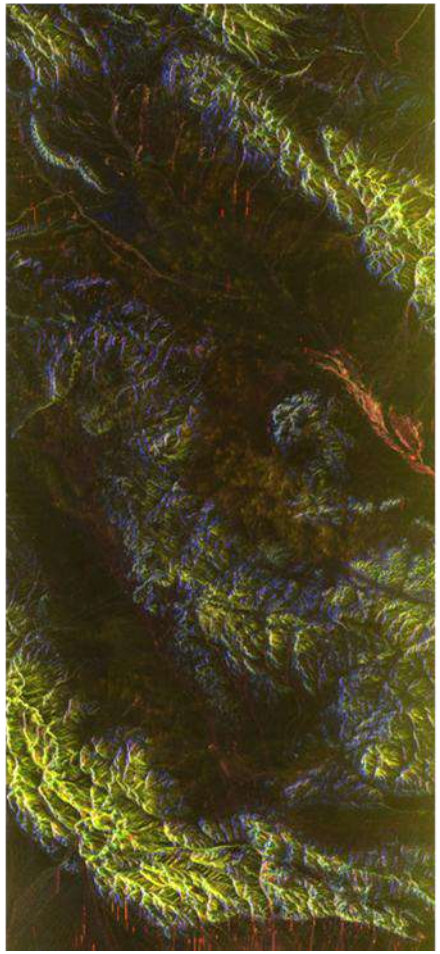


L1A Prev #2



Geology

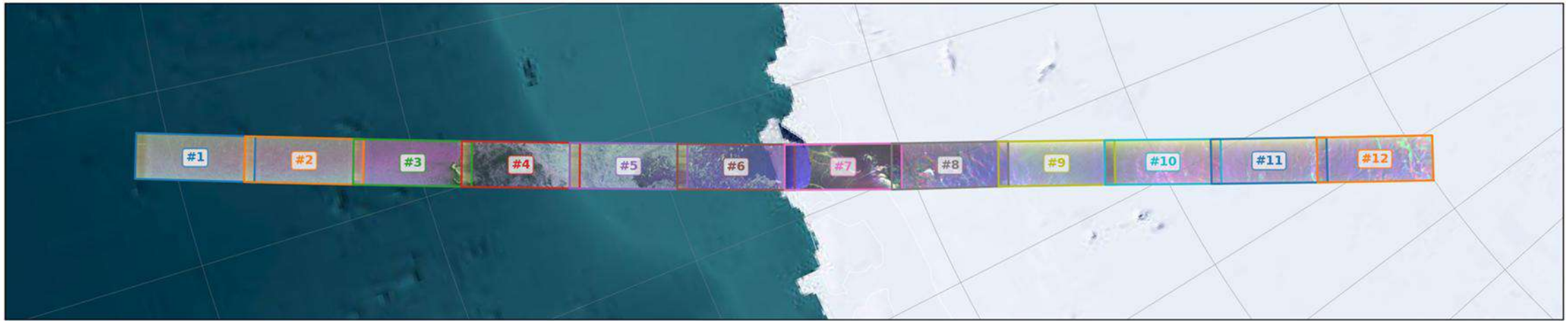
PRODUCT LEVEL	DATA TAKE ID
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PRODUCT LEVEL	DATA TAKE ID
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Antarctica & Ocean



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BIO_S3_SCS__1S_20260119T004127_20260119T004147_T_G01_M01_C06_T030_F216_01_DLHA7H
BIO_S3_SCS__1S_20260119T004146_20260119T004206_T_G01_M01_C06_T030_F217_01_DLHA68
BIO_S3_SCS__1S_20260119T004205_20260119T004225_T_G01_M01_C06_T030_F218_01_DLHA6Q
BIO_S3_SCS__1S_20260119T004224_20260119T004245_T_G01_M01_C06_T030_F219_01_DLHA9K
BIO_S3_SCS__1S_20260119T004243_20260119T004304_T_G01_M01_C06_T030_F220_01_DLHA70
BIO_S3_SCS__1S_20260119T004302_20260119T004323_T_G01_M01_C06_T030_F221_01_DLHACN
BIO_S3_SCS__1S_20260119T004321_20260119T004342_T_G01_M01_C06_T030_F222_01_DLHA83
BIO_S3_SCS__1S_20260119T004340_20260119T004401_T_G01_M01_C06_T030_F223_01_DLHA C6
BIO_S3_SCS__1S_20260119T004359_20260119T004420_T_G01_M01_C06_T030_F224_01_DLHA8S
BIO_S3_SCS__1S_20260119T004418_20260119T004439_T_G01_M01_C06_T030_F225_01_DLHAEB